

**NUMERICAL INVESTIGATION OF SURFACE
GEOMETRY EFFECTS ON THE
AEROTHERMODYNAMICS OF A HYPERSONIC
INFLATABLE AERODYNAMIC DECELERATOR**

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APPROVAL PAGE

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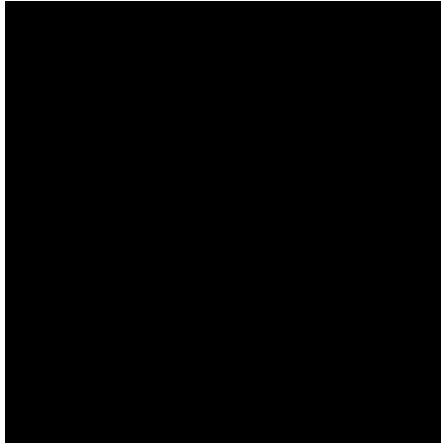
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ABSTRACT

This thesis presents a comprehensive investigation into the aerothermal characteristics of Hypersonic Inflatable Aerodynamic Decelerators (HIADs), with a particular focus on the impact of their scalloped surface topology on flow physics and aerodynamic heating. Utilizing Computational Fluid Dynamics (CFD), the study reviews related research in HIAD development and re-entry aerothermodynamics. It delves into the fundamental principles of hypersonic aerodynamics, including shock wave formation, viscous interactions, and critical heat transfer mechanisms. The work also compares conventional Thermal Protection Systems (TPS) with flexible HIAD structures, highlighting the benefits and challenges of the latter. Furthermore, the theoretical framework of CFD, turbulence modeling (specifically the SST k-omega model), and advanced thermodynamic gas models are discussed to establish a robust methodology for analyzing complex hypersonic flow fields and predicting localized peak heating effects.

Keywords: Hypersonic Inflatable Aerodynamic Decelerator (HIAD), Aerothermodynamics, Computational Fluid Dynamics (CFD), Shock-Boundary Layer Interaction (SBLI), Scalloped Geometry, Aerodynamic Heating, Thermally Perfect Gas

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LIST OF SYMBOL

Notation	Definition
M	Mach Number
l	Characteristic line
α	Characteristic Angle
V	Prandtl-Meyer Angle
μ	Mach Angle
S	Approximate Cross-Sectional Area
r	Radius
e	Exit
A	Area
γ	Ratio of Specific Heat
$\in$	Area Ratio
θ	Expansion Wave Angle
ρ	Density
u	Velocity
P	Pressure
h	Enthalpy
$*$	Critical
T	Temperature
P	Pressure
F	Force
$\dot{m}$	Mass Flow Rate
I_{SP}	Specific Impulse

LIST OF ABBREVIATION

SYMBOL	Definition
CFD	Computational Fluid Dynamics
RANS	Reynolds Averaged Navier Stokes
SST	Shear Stress Transport
RMS	Root Mean Square
2D	Two - Dimensional
3D	Three - Dimensional
SYM	Symmetry
GDT	Grid Dependency Test
PR	Pressure Ratio

CHAPTER 1

INTRODUCTION

1.1 Background

In this day and age, Exploration of space has been one of the driving force for technological innovation, pushing the boundaries on what is possible in the human. There's a challenge in space exploration is the safe return of spacecraft and payloads to Earth or other celestial bodies with an atmosphere. During atmospheric reentry, a vehicle traveling at hypersonic speeds experiences extreme aerodynamic heating, which can lead to catastrophic failure if not properly managed. To protect the vehicle from these extreme temperatures, a thermal protection system (TPS) is required.

For generations, Rigid-like TPS that have been the standard solution for atmospheric earth reentry such as Reusable TPS, Structurally Integrated TPS, Hot Structure, and Ablator (Bouslog, 2023). However, the size of a rigid heat shield is limited by the launch vehicle's payload fairing, which in turn restricts the size and mass of the reentry vehicle. This limitation poses a challenge for future missions that require the landing of large payloads and its flexibility on its backups, such as human missions to Mars or the return of large assets from space.

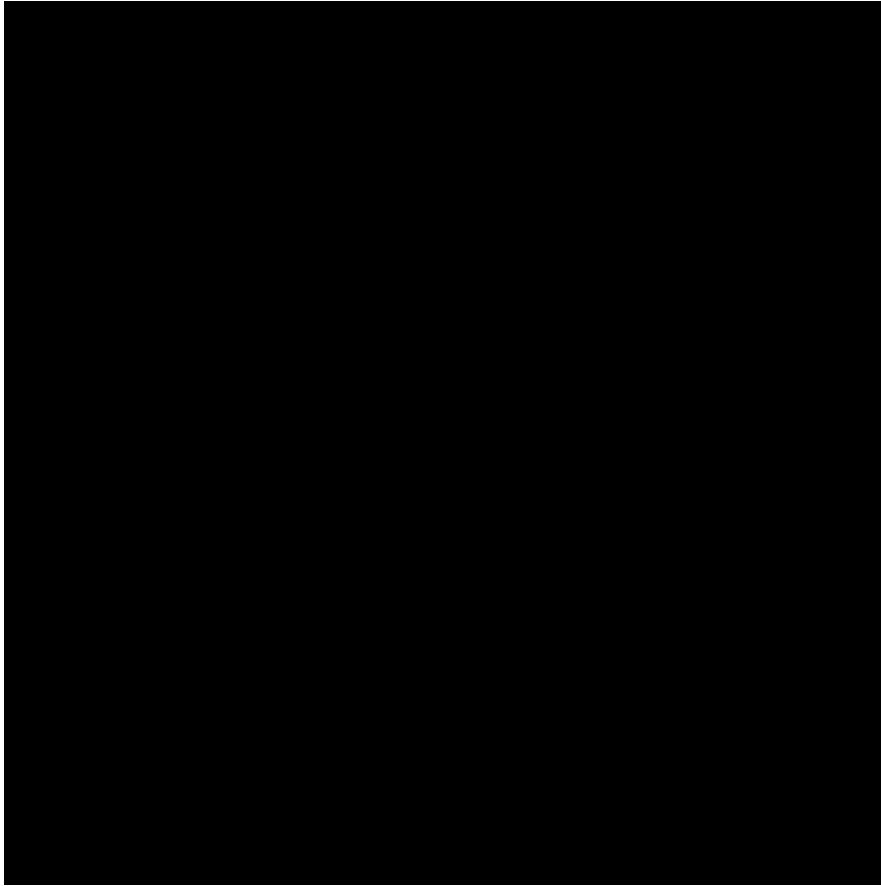


Figure 1.1: Types of Thermal Protection Shield/System

To counteract this challenge, an deployable entry systems called inflatable aerodynamic decelerators (IADs) TPS has been emerged and tested as a non integrated TPS and flexible (DiNonno & Cheatwood, 2024). IADs, also known as hypersonic inflatable aerodynamic decelerators (HIADs), are lightweight, flexible structures that can be packed into a small volume during launch and then inflated to a much larger size before atmospheric entry. The large surface area of an IAD provides significant aerodynamic drag, allowing for the deceleration of large masses at higher altitudes where the air is less dense and the heating is less severe.



Figure 1.2: HIADs when deployed

The development of IADs approach that combines aerodynamics, structural mechanics, and materials science. A key component of an IAD is the flexible thermal protection system (FTPS), which is responsible for protecting the inflatable structure from the intense heat of reentry. The FTPS is a multi-layered system that typically consists of an outer fabric, insulation layers, and a gas barrier (Guo et al., 2011). The design and analysis of the FTPS task that requires accurate modeling of the aerothermal loads and the material response.

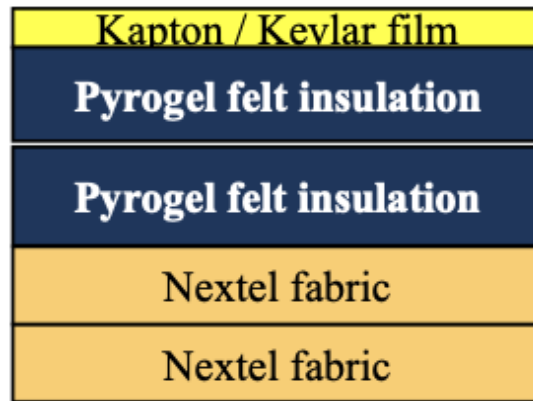


Figure 1.3: IADs Layer Example

1.2 Problem Statement

The successful design and operation of an IAD depend on the ability to accurately predict and manage the unusual edge aerothermodynamic environment and the complex fluid-structure interaction (FSI) phenomena that occur during atmospheric reentry. The flexibility of the IAD structure and the FTPS adds a layer of complexity to the analysis

that is not present in traditional rigid heat shields (Missing Citation). The deformation of the IAD under aerodynamic loads can affect the flow field and the heat transfer to the vehicle, which in turn can affect the structural response.

Therefore, there is a critical need for numerical tools and methodologies to model the coupled FSI and aerothermal behavior of IADs. These tools must be able to capture the complex physics of hypersonic flows, including real gas effects, chemical reactions, and turbulence. They must also be able to model the nonlinear structural response of the inflatable structure and the thermal response of the FTPS.

1.3 Research Objective

In this research of this thesis, the objectives is to develop and to study a computational fluid dynamics (CFD)-based methodology for the aerothermal and aerodynamic analysis of a flexible thermal protection system for an inflatable aerodynamic decelerator (IAD). The study will focus on adapting and applying the methodology presented in the work of Al-Obaidi et al. (Al-Obaidi, Al-Obaidi, & Al-Obaidi, 2021) to the specific case of an IAD.

The specific objectives are:

1. **Geometric Modeling:** To generate 2D axisymmetric CFD models of a Hypersonic Inflatable Aerodynamic Decelerator (HIAD) based on IRVE-3 specifications, specifically contrasting the *scalloped* surface topology/geometry of stacked toroids against an equivalent *smooth* rigid sphere-cone baseline.
2. **Flow Field Geometrical Model Analysis:** To analyze the impact of surface waviness on the hypersonic shock structure, specifically quantifying the *bow shock standoff distance* (Δ) and identifying the presence and stability of *recirculation zones* (vortices) within the gaps (valleys) of the inflatable rings.
3. **Aerothermal Load Analysis:** To determine and compare the surface distributions of *Wall Heat Flux* (q_w) and *Wall Shear Stress* (τ_w), with a focus on identifying peak heating augmentation at the toroidal crests versus heat flux alleviation in the valleys compared to the smooth baseline.
4. **Aerodynamic Performance Evaluation:** To calculate and compare the integral

aerodynamic forces, specifically the *Drag Coefficient* (C_d), to determine if the scalloped geometry provides a deceleration advantage over the rigid baseline.

1.4 Scope and Limitations

The scope of this research is strictly defined to provide aerothermal and aerodynamic analysis of the Flexible Thermal Protection System (FTPS) within the constraints of available computational resources. The boundaries and limitations of this study are as follows:

1. **Geometric Representation:** The HIAD structure is simplified by assuming a fully inflated and rigid shape. While the actual inflatable decelerator is flexible, this study treats the "scalloped" toroidal stack as a non-deformable geometry to isolate the impact of surface topology on the flow field.
2. **Dimensionality:** The investigation is restricted to a **2D Axisymmetric** model. This approach assumes a zero angle of attack ($\alpha = 0^\circ$) to maintain rotational symmetry, allowing for higher grid resolution in the boundary layer while remaining within hardware memory (RAM) limits.
3. **Numerical Framework:** The fluid flow is governed by the *Reynolds-Averaged Navier-Stokes* (RANS) equations. The simulations are executed using the *Density-Based Coupled Solver* within the ANSYS Fluent commercial numerical suite, utilizing a second-order upwind discretization scheme for shock capturing.
4. **Turbulence Modeling:** The *Shear Stress Transport* (SST) $k - \omega$ model is employed to resolve the turbulence. This model is specifically selected for its superior ability to handle flow separation and recirculation zones within the toroidal valleys.
5. **Thermophysical Gas Model:** To account for the extreme hypersonic conditions, *real gas effects* are incorporated through a **Thermally Perfect Gas** model, which accounts for the temperature-dependency of specific heats (C_p).
6. **Chemical Modeling:** Although the flow is hypersonic, chemical dissociation is handled through a simplified approach. While the high-enthalpy effects are acknowledged, the gas is assumed to be in *chemical equilibrium* or frozen, neglect-

ing finite-rate non-equilibrium chemical reactions. This limitation is necessary to ensure solver stability and computational efficiency on standard workstation hardware.

1.5 Methodology Overview

This study approach using simulation-based to analyze the aerothermal performance of surface topology/geometry on a Hypersonic Inflatable Aerodynamic Decelerator (HIAD). The data used in this study is derived from parametric geometric modeling then resolving flow gradients such as shock waves and boundary layers. From this analysis, the scalloped (inflatable) geometry is compared against a smooth (rigid) baseline to evaluate its thermal mitigation potential. The study begins with a comprehensive literature review and concludes with an estimation of aerodynamic heating reduction, as illustrated in Figure 1.4. The research stages are structured sequentially as follows:

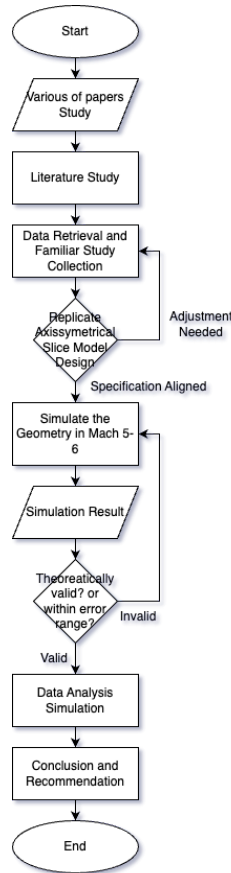


Figure 1.4: Research Methodology Flowchart.

a. Literature Study

The initial stage of the study is conducting literature study. In this phase, the researcher reviews various references related to the concept of Hypersonic Aerothermodynamics, the physics of shock-boundary layer interaction, the implementation of turbulence modeling (SST $k - \omega$), and the formulation of Thermally Perfect Gas models. This literature review is conducted to obtain a strong theoretical foundation for designing the computational domain, determining the critical flow parameters (Ma , Re , T), and supporting the analysis and simulation stages.

b. Geometric Modeling and Data Generation

Research data is obtained from parametric geometric modeling generated via Python scripting within the FreeCAD environment. This data contains the coordinate definitions of the vehicle profiles, specifically the toroidal radii and valley depths for the "scalloped" configuration and the tangent-cone angle for the "smooth" baseline. These geometric parameters serve as the primary input for the discretization process.

c. Numerical Simulation and Processing

This stage encompasses several key computational steps. First, the geometric data is discretized into a hybrid mesh with strict near-wall refinement ($y^+ \approx 1$) to resolve the viscous sublayer. Next, the physical models are configured within the Density-Based Coupled Solver, specifically enabling the Equation of State for compressible flow and Sutherland's Law for viscosity.

The simulation process aims to resolve the flow field around the vehicle to identify regions of peak heating and flow separation. The meshed geometries are subjected to hypersonic freestream conditions ($Ma > 5$) to represent the re-entry environment. The simulation is conducted using a steady-state approach to determine the converged values of heat flux and drag.

The initial simulation results are validated against theoretical correlations. If significant discrepancies are found (e.g., residuals not converging or physical instability), the grid parameters are refined, and the simulation is repeated until the solution achieves statistical stationarity. Thus, the simulation not only yields flow visualization but also ensures the numerical validity of the aerothermal predictions.

d. Data Analysis

In this stage, the simulation results are analyzed to evaluate the thermal efficiency of the proposed geometry. The analysis is performed by comparing two main scenarios: the scalloped inflatable profile and the smooth rigid baseline. The parameters compared include the Stagnation Point Heat Flux, the Wall Heat Flux distribution along the flank, and the total Drag Coefficient (C_d). The results are then used to calculate the percentage reduction in peak heating and to identify any local hot spots caused by shock impingement in the valleys.

e. Conclusion and Recommendations

The final stage of the research is the current conclusions and what comes next. Based on the analysis results, the thermal performance of the scalloped geometry is determined relative to the baseline. This research also presents recommendations regarding the optimal geometric configuration (toroid radius vs. gap size) as an empirical basis to support the design of future systems. Recommendations are addressed to future researchers and aerospace engineers as a basis for decision-making in the effort to improve the efficiency and thermal safety of atmospheric re-entry vehicles.

1.6 Thesis Outline

The structure of this thesis arranged as such. A brief description of each chapter is outlined as follows:

CHAPTER I This chapter contains the research background, problem statement, research objectives, scope of study, methodology overview, and the systematic thesis outline. This section is providing an initial overview regarding the rationale for conducting the research, the specific goals to be achieved, and the limitation of the research scope, and established the writing framework/systematic plan.

CHAPTER II This chapter contains the theoretical framework and literature review that supports this. It discusses the fundamental concepts of Hypersonic Aerothermodynamics, the physics of Shock-Boundary Layer Interaction (SBLI), the formulation of the Shear Stress Transport (SST) turbulence model, and the thermodynamic properties of high-temperature gases (Thermally Perfect Gas). This chapter also reviews relevant previous studies to establish the state-of-the-art.

CHAPTER III This chapter contains the research schedule plan, covering the sequential stages of the investigation. This includes the parametric geometric modeling in FreeCAD, the computational domain discretization (meshing strategy), the numerical configuration of the density-based CFD solver, and the validation procedures. It describes the complete workflow of the research execution from the initial design to the final simulation.

CHAPTER IV This chapter contains the theoretical hypothesis and predicted aerothermal phenomena based on the literature study. Instead of final simulation results, this chapter synthesizes existing research to predict/forecast the expected result characteristics, such as shock wave structures and recirculation regions in the scalloped valleys. It establishes the performance expectations and physical hypotheses that will be subsequently verified during the simulation phase.

CHAPTER V contains the summary of the research findings derived from the data analysis in the previous chapter. This section presents the final conclusions regarding the aerothermal performance of the scalloped HIAD and provides recommendations for future researchers and aerospace engineers as a basis for further development in hypersonic deceleration systems.

CHAPTER 2

Literature Review

2.1 Review of Related Research

To establish a baseline for the expected aerothermal outcomes, a review of key flight tests and numerical studies was conducted. Table 2.1 summarizes the landmarks in HIAD research that support the triangulation of results in this study.

Table 2.1: Comparison of Landmark HIAD Aerothermal Research and Flight Tests

Research Project	Key Geometry	Major Findings	Aerothermal Outcome
NASA IRVE-3 (Lau et al., 2013)	3-meter stacked-toroid HIAD.	Confirmed that "scallop-ing" (surface deflection) increases local heating compared to smooth surfaces.	Primary validation for the scalloped model.
NASA LOFTID (Cassell et al., 2022)	6-meter large-scale HIAD with Gen-2 FTPS.	Demonstrated thermal survival during Earth reentry; identified transition onset on the flank.	Validates FTPS effectiveness at high enthalpy.
Lichodziejewski et al. (2012)	Deformed Outer Mold Line (OML) Analysis.	Proved that surface irregularities between tori significantly augment local heat flux.	Justifies the Config B (Smooth) comparison.
HEART Project (Cassell et al., 2014)	Optimized Nose-IAD Junction Geometry.	Found that the juncture between the nose and the IAD creates a secondary heat flux peak.	Predicts the location of local hot spots.
Continued on next page			

Table 2.1 – continued from previous page

Research Project /	Key Geometry	Major Findings	Aerothermal	Relevance to Outcome
Menter (1994)	<i>k</i> - ω SST Turbulence Model.	Superior performance in predicting separation in adverse pressure gradients.		Supports model for valley recirculation.

2.1.1 Triangulation of Expected Outcomes

By synthesizing the findings of *Lau et al. (2013)* and *Lichodziejewski et al. (2012)*, it is hypothesized that the scalloped geometry in this study will generate localized heat flux peaks on the toroidal crests.

Furthermore, the implementation of the *SST k – ω* turbulence model, as validated by *Menter (1994)*, is expected to resolve the stable recirculation cells within the toroidal valleys. Finally, the use of a *Thermally Perfect Gas* model reflects the high-enthalpy conditions observed in the *LOFTID* mission, ensuring that the predicted shock physics align with real-gas observations rather than simplified ideal-gas assumptions.

This synthesis establishes the theoretical foundation for the numerical methodology described in **Chapter 3**.

2.2 Fundamentals of Hypersonic Aerodynamics

Hypersonic flow is commonly as a rule of thumb being defined as the regime where the Mach number exceeds five ($M > 5$), though this threshold is more physical than mathematical. As a vehicle’s velocity increases to several kilometers per second, the kinetic energy of the flow becomes so vast that its conversion into internal energy across shock waves and within viscous boundary layers fundamentally alters the gas behavior.

According to the classical frameworks established in hypersonic theory (Citation Needed), the flowfield is characterized by five distinct physical phenomena that distinguish it from supersonic flow:

1. **Thin Shock Layers:** At high Mach numbers, the shock wave lies very close to the body surface. For a blunt body, this results in a small stand-off distance, creating a high-pressure region between the shock and the vehicle.

2. **Entropy Layers:** The strong curvature of the bow shock at the nose creates a region of high entropy gradient. As this layer flows downstream, it interacts with the boundary layer, complicating the pressure and heat transfer calculations.
3. **Viscous Interaction:** The high temperature within the boundary layer reduces the local density, causing the boundary layer to grow rapidly ($\delta \propto M^2/\sqrt{Re}$). This thick layer displaces the outer inviscid flow, significantly altering the surface pressure distribution.
4. **High-Temperature Effects:** The extreme kinetic energy is dissipated into thermal energy, triggering vibrational excitation, dissociation, and even ionization of the air molecules. This renders the calorically perfect gas assumption ($c_p = \text{const.}$) invalid.
5. **Low-Density Effects:** At very high altitudes, the continuum assumption may begin to break down as the mean free path of the molecules becomes comparable to the vehicle's characteristic length.

For Inflatable Aerodynamic Decelerators (IADs), understanding these fundamentals is critical. The "scaloped" surface topology of stacked toroids interacts directly with the thin shock layer and viscous boundary layer, creating complex flow structures that do not exist on smooth-body vehicles.

2.2.1 Mach Number and the Hypersonic Regime ($Ma > 5$)

The transition from supersonic to hypersonic flow is generally defined by the rule of thumb $Ma > 5$. Unlike the transition from subsonic to supersonic flow ($Ma = 1$), which is marked by a distinct physical discontinuity (the formation of shock waves), the boundary for hypersonic flow is not physically discrete. Instead, $Ma \approx 5$ represents the threshold where certain physical approximations used in linear supersonic theory lose accuracy, and high-temperature gas dynamics become significant (Anderson, 2006).

The distinguishing characteristic of this regime is the dominance of flow kinetic energy over internal energy. From the steady-flow energy equation, the total enthalpy h_0 is conserved across the shock wave:

$$h_0 = h_\infty + \frac{1}{2}V_\infty^2 = h + \frac{1}{2}V^2 \quad (2.1)$$

Where h is the static enthalpy and V is the velocity. In the hypersonic regime, the freestream velocity V_∞ is extremely high, meaning the kinetic energy term $\frac{1}{2}V_\infty^2$ is significantly larger than the static enthalpy h_∞ . As the flow decelerates to stagnation conditions ($V \approx 0$) at the nose of a blunt body (such as an IAD), this massive kinetic energy is converted almost entirely into internal energy (heat).

This energy conversion scales with the square of the Mach number. For a calorically perfect gas, the ratio of stagnation temperature T_0 to freestream temperature T_∞ is given by:

$$\frac{T_0}{T_\infty} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \quad (2.2)$$

For a flight at $Ma = 5$ in standard air ($\gamma = 1.4$), the temperature rises by a factor of 6. At $Ma = 10$, it rises by a factor of 21. This exponential increase in temperature necessitates the coupling of thermodynamics with aerodynamics (aerothermodynamics), as the assumption of constant specific heats (C_p) is no longer valid due to vibrational excitation and dissociation of gas molecules.

2.2.2 Shock Wave Formation and Properties

In the hypersonic regime, the vehicle moves through the atmosphere significantly faster than the speed of sound, preventing pressure disturbances from propagating upstream. Consequently, these disturbances coalesce ahead of the vehicle to form a strong shock wave, a thin region of abrupt discontinuity in fluid properties (Anderson, 2006).

The thermodynamic changes across a normal shock wave are governed by the conservation of mass, momentum, and energy. These are collectively known as the Rankine-Hugoniot relations. For a calorically perfect gas, the ratio of downstream (post-shock, subscript 2) to upstream (pre-shock, subscript 1) properties is a function of the normal Mach number component (M_n):

$$\frac{p_2}{p_1} = 1 + \frac{2\gamma}{\gamma + 1} (M_n^2 - 1) \quad (2.3)$$

$$\frac{T_2}{T_1} = \left[1 + \frac{2\gamma}{\gamma + 1} (M_n^2 - 1) \right] \left[\frac{2 + (\gamma - 1)M_n^2}{(\gamma + 1)M_n^2} \right] \quad (2.4)$$

A critical distinction in hypersonic aerodynamics is the behavior of these ratios as the Mach number approaches infinity ($M \rightarrow \infty$). While pressure and temperature ratios increase indefinitely (leading to extreme aerodynamic heating), the density ratio approaches a finite limit:

$$\lim_{M \rightarrow \infty} \frac{\rho_2}{\rho_1} = \frac{\gamma + 1}{\gamma - 1} \quad (2.5)$$

For standard air ($\gamma = 1.4$), this limiting density ratio is $\rho_2/\rho_1 \approx 6$. This high compression means the mass flow is squeezed into a much smaller volume behind the shock compared to supersonic flow. This phenomenon is physically responsible for the formation of the Thin Shock Layer, where the shock wave "wraps" very closely around the body surface. This proximity is particularly important for the scalloped geometry of an IAD, as the shock structure may impinge upon the peaks of the toroidal rings, creating localized hotspots.

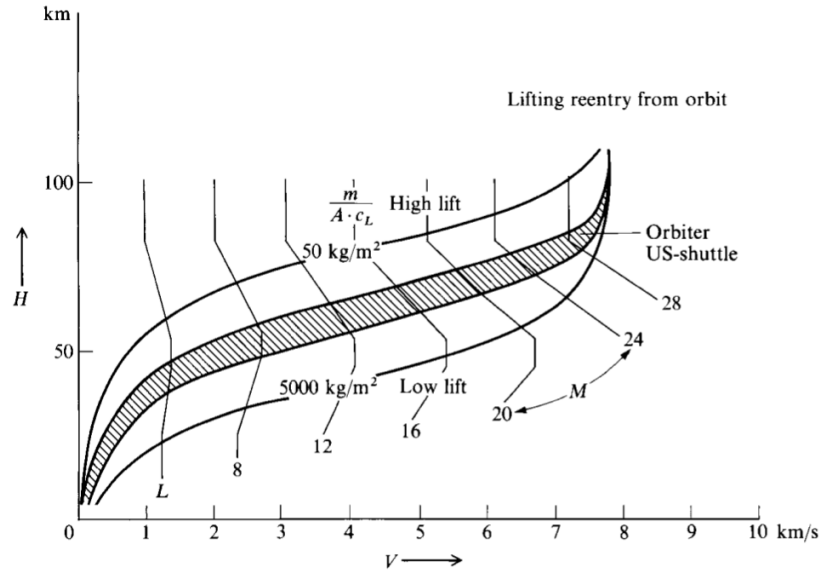


Figure 2.1: Figure for Shock Wave Formation and Properties Relation (Anderson, 2006)

2.2.3 Invalidity of Subsonic/Supersonic Assumptions (Entropy layers, Viscous interaction)

In lower-speed supersonic flows, aerodynamic analysis often decouples the inviscid outer flow from the viscous boundary layer. However, in the hypersonic regime ($Ma >$

5), this decoupling is often invalid due to two dominant phenomena: the Entropy Layer and Viscous Interaction (Anderson, 2006).

The Entropy Layer

For blunt bodies like an Inflatable Aerodynamic Decelerator (IAD), the bow shock is highly curved. The shock strength varies significantly from the normal shock at the stagnation streamline (strongest) to the oblique shock at the flanks (weaker). According to Crocco's Theorem, this curvature generates a strong gradient of entropy (Δs) across the streamlines behind the shock.

This creates a layer of high-entropy, vortical flow—termed the "entropy layer"—that flows downstream and wets the vehicle surface. Unlike in supersonic flow, where this layer is negligible, in hypersonic flow, the entropy layer can be comparable in thickness to the boundary layer. This invalidates the standard boundary layer assumption that flow properties at the edge of the boundary layer are uniform/irrotational.

Viscous Interaction

A fundamental assumption in classical Prandtl boundary layer theory is that the pressure gradient is impressed upon the boundary layer by the outer inviscid flow (i.e., $\partial p / \partial y \approx 0$). In hypersonic flow, the massive kinetic energy converted to heat causes the temperature within the boundary layer to rise significantly. Since pressure is constant across the layer, the density decreases inversely with temperature ($\rho \propto 1/T$), causing the boundary layer displacement thickness (δ^*) to grow rapidly:

$$\delta^* \propto \frac{M_\infty^2}{\sqrt{Re_x}} \quad (2.6)$$

This thick boundary layer acts as a new "effective" shape for the vehicle, displacing the outer inviscid flow and generating a completely different pressure distribution than what the geometric shape alone would suggest. This mutual dependence—where the boundary layer determines the pressure field, and the pressure field determines the boundary layer growth—is known as Hypersonic Viscous Interaction. For the scalloped geometry of the IAD, this interaction is critical in the valleys between toroids, where thick boundary layers may bridge the gaps and alter the effective aerodynamic shape.

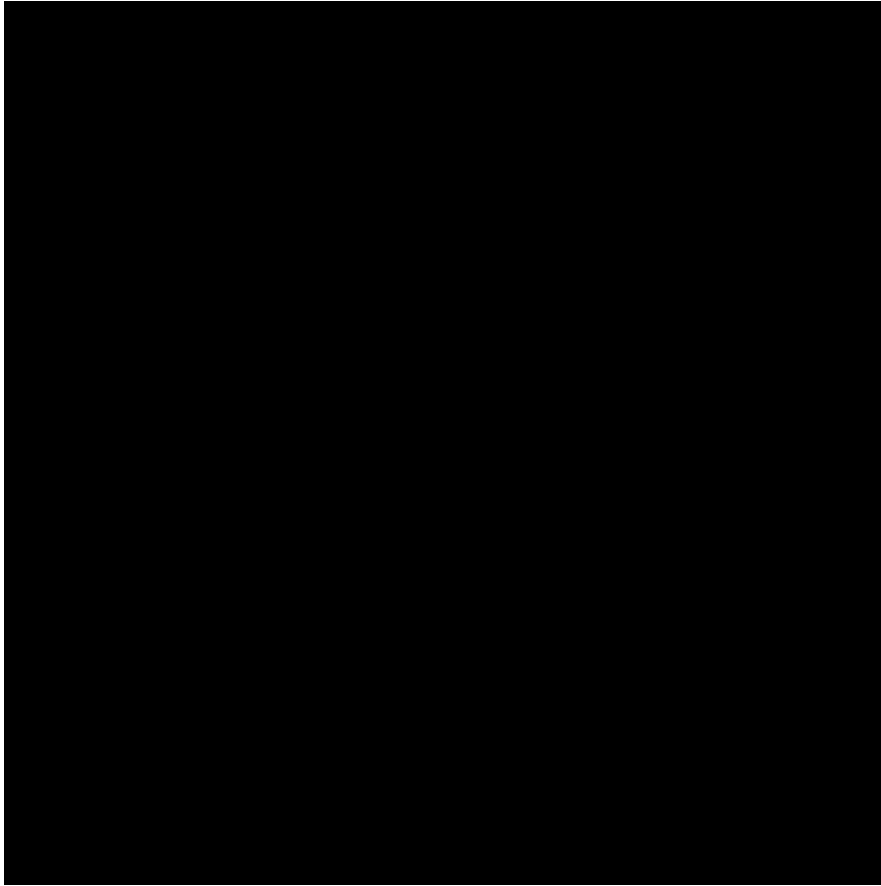


Figure 2.2: Figure for Invalidity of Subsonic/Supersonic Assumptions.

2.3 Aerothermodynamics of Re-entry Vehicles

Aerothermodynamics represents the synthesis of high-speed aerodynamics and high-temperature thermodynamics, forming the cornerstone of atmospheric re-entry vehicle design. Unlike conventional aerodynamics, where the fluid is treated as a continuum with constant chemical properties, aerothermodynamics must account for the intense coupling between the flow field and the state of the gas (Anderson, 2006).

The fundamental challenge of re-entry is the dissipation of orbital energy. A spacecraft returning from Low Earth Orbit (LEO) possesses a velocity of approximately 7.8 km/s, while vehicles returning from lunar or Martian trajectories can exceed speeds of 11 km/s (Anderson, 2006). To reach the surface safely, this immense kinetic energy ($\frac{1}{2}mV^2$) and potential energy (mgh) must be converted into other forms—primarily thermal energy—without exceeding the structural limits of the vehicle or the physiological

limits of the payload.

This energy conversion occurs within the shock layer enveloping the vehicle. As the gas molecules are processed by the strong bow shock, their directed kinetic energy is randomized into internal energy, raising the static temperature of the shock layer to values often exceeding the temperature of the solar surface (> 6000 K). At these extreme temperatures, the air acts as a chemically reacting plasma rather than a simple gas. The nitrogen and oxygen molecules dissociate ($N_2 \rightarrow 2N$, $O_2 \rightarrow 2O$) and, at higher speeds, ionize ($N \rightarrow N^+ + e^-$), significantly altering the thermodynamic and transport properties of the fluid (e.g., specific heat ratio γ , viscosity μ , and thermal conductivity k).

Consequently, the design of any re-entry system, including Hypersonic Inflatable Aerodynamic Decelerators (HIADs), is governed by the need to manage this heat load. The vehicle must be designed not only to withstand the integrated heat load (total heat soaked) but also the peak heat flux (maximum rate of heating) which occurs deep within the atmosphere where density increases. This creates a critical trade-off between aerodynamic deceleration (drag) and thermal management, dictating that the vehicle geometry must be optimized to maximize drag while minimizing the transfer of energy to the surface.

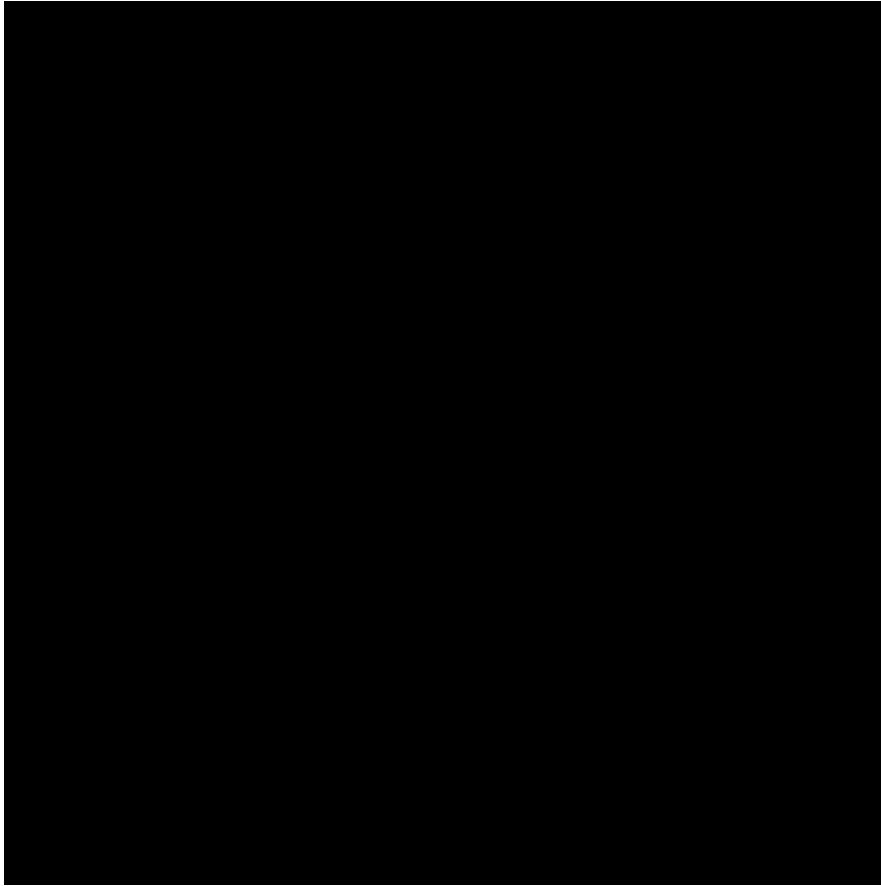


Figure 2.3: Figure for Aerothermodynamics of Re-entry Vehicles.

2.3.1 Mechanisms of Aerodynamic Heating (Kinetic to Thermal energy conversion)

The primary source of aerodynamic heating in hypersonic flight is the immense kinetic energy of the oncoming flow. As the vehicle arrests the flow, this directed kinetic energy is converted into internal thermal energy. This conversion occurs through two distinct physical mechanisms within the shock layer (Anderson, 2006).

1. Shock Wave Compression

As the supersonic freestream crosses the bow shock, the flow velocity decreases abruptly, and the pressure and density rise. This process is largely adiabatic but highly non-isentropic. The work done by pressure forces on the fluid elements compresses the gas, raising the static enthalpy (h) significantly. At the stagnation point, effectively

all the freestream kinetic energy is converted to enthalpy, defined by the total enthalpy equation:

$$h_0 = h_\infty + \frac{1}{2}V_\infty^2 \quad (2.7)$$

For a vehicle traveling at $Ma = 25$ (approximate re-entry speed), the total enthalpy is sufficient to dissociate nitrogen and oxygen, creating a chemically reacting plasma behind the shock.

2. Viscous Dissipation within the Boundary Layer

While the shock wave processes the bulk flow, the interaction at the vehicle surface is governed by viscosity. Within the boundary layer, the fluid velocity must vanish at the wall ($u = 0$) due to the no-slip condition. This deceleration is caused by shear stress (τ) acting between adjacent fluid layers. The work done by these viscous forces against the velocity gradient generates heat, a phenomenon known as viscous dissipation.

The local rate of heat transfer to the wall (q_w) is determined by the thermal gradient at the surface, governed by Fourier's Law:

$$q_w = -k_w \left(\frac{\partial T}{\partial y} \right)_w \quad (2.8)$$

In hypersonic aerothermodynamics, it is convenient to express this heat flux in terms of the driving enthalpy potential. The heat flux is proportional to the difference between the recovery enthalpy (h_{aw} , the enthalpy of the gas if the wall were adiabatic) and the actual wall enthalpy (h_w):

$$q_w = \rho_e u_e C_H (h_{aw} - h_w) \quad (2.9)$$

Where ρ_e and u_e are the edge density and velocity, and C_H is the Stanton number (heat transfer coefficient). This relationship highlights that heating is most severe where the boundary layer is thinnest (high gradients) and where the local flow energy is highest.

2.3.2 Bow Shock Stand-off/detachment Distance

A defining characteristic of blunt-body aerodynamics in the hypersonic regime is the detachment of the shock wave from the vehicle's nose. Unlike sharp-edged bodies where the shock may be attached to the leading edge, the large radius of curvature of an Inflatable Aerodynamic Decelerator (IAD) forces the formation of a strong, curved normal shock ahead of the stagnation point. The physical separation between the shock wave and the body along the stagnation streamline is defined as the shock stand-off distance (Δ).

The magnitude of this stand-off distance is critically dependent on the density ratio across the shock. As derived in classical hypersonic theory (Anderson, 2006), the conservation of mass requires that the mass flow entering the shock must exit through the annular area between the shock and the body. Because the post-shock density (ρ_2) is significantly higher than the freestream density (ρ_∞), the required flow area is small, resulting in a thin shock layer.

For a spherical nose of radius R , the stand-off distance can be approximated by the correlation:

$$\frac{\Delta}{R} \approx \frac{\rho_\infty}{\rho_2} \quad (2.10)$$

More precise empirical correlations, such as Billig's approximation for a sphere, express this relationship as:

$$\frac{\Delta}{R} = 0.143 \exp\left(\frac{3.24}{M_\infty^2}\right) \quad (2.11)$$

This relationship has profound implications for IAD design. In hypersonic flow, where real gas effects (dissociation) can increase the density ratio ρ_2/ρ_∞ well beyond the ideal gas limit of 6, the stand-off distance shrinks further. A smaller Δ increases the velocity gradients within the shock layer, which directly intensifies the convective heat transfer to the wall. For scalloped geometries, if the stand-off distance becomes comparable to the depth of the valleys between toroids, it can lead to complex shock-shock interactions or shock impingement on the structural surface.

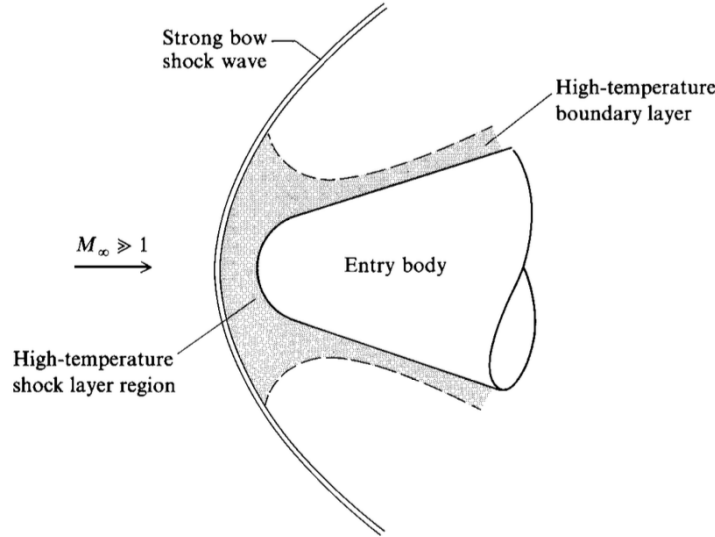


Figure 2.4: Figure for Bow Shock Stand-off/detachment Distance. (Anderson, 2006)

2.3.3 H. Julian Allen's Blunt Body Theory

In the early development of supersonic flight, the prevailing design philosophy favored sharp, slender noses to minimize aerodynamic drag. However, as speeds approached the hypersonic regime, H. Julian Allen and Alfred Eggers of NACA Ames Research Center revolutionized re-entry vehicle design with the Blunt Body Theory (1953) (Anderson, 2006). This counter-intuitive theory established that while sharp bodies are efficient for supersonic cruise, they are catastrophic for hypersonic re-entry.

The fundamental principle rests on the energy balance of the vehicle. The total kinetic energy lost by the re-entering body is dissipated into two sinks: heating the air (via shock compression) and heating the vehicle body (via viscous friction).

1. **Sharp Body:** Creates a weak, attached shock wave. The wave drag is low, meaning the air is not heated significantly. Consequently, a large fraction of the kinetic energy is dissipated through skin friction directly on the surface, leading to extreme aerodynamic heating that can melt conventional materials.
2. **Blunt Body:** Generates a strong, detached bow shock wave. This creates massive pressure drag (wave drag), which dissipates the majority of the kinetic energy into the airflow itself, creating a high-temperature wake behind the vehicle. Because

the energy is dumped into the atmosphere rather than the structure, the heat load on the vehicle is drastically reduced.

Allen mathematically demonstrated that the stagnation point heat flux (q_s) is inversely proportional to the square root of the nose radius (R_n), while the aerodynamic drag (D) is directly proportional to the projected area (and thus R_n):

$$q_s \propto \frac{1}{\sqrt{R_n}} \quad \text{and} \quad D \propto R_n^2 \quad (2.12)$$

This relationship proves that increasing the bluntness of the vehicle not only increases drag (aiding deceleration) but simultaneously decreases the peak heat flux. This theorem is the theoretical foundation for Hypersonic Inflatable Aerodynamic Decelerators (HIADs), which utilize an extremely large effective radius to maximize drag and minimize localized heating, thereby enabling the entry of massive payloads into planetary atmospheres.

Here is example of the effect of the blunt body (Hamza, Khan, & Maqsood, 2022)

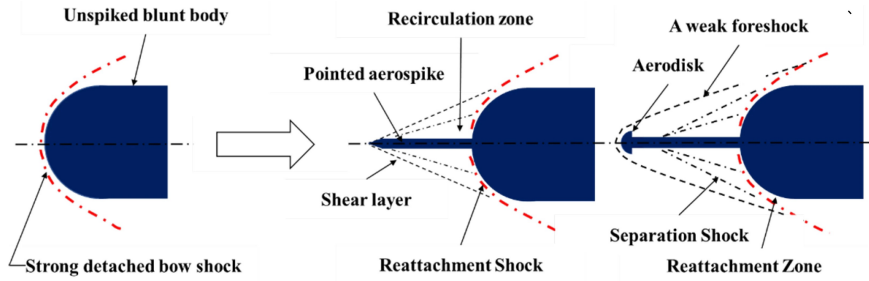


Figure 2.5: Figure for H. Julian Allen's Blunt Body Theory.

2.4 Thermal Protection Systems (TPS) Technology

The Thermal Protection System (TPS) is the single most critical subsystem for atmospheric entry vehicles, serving as the interface between the extreme aerothermal environment and the vehicle's structure. Its primary function is to maintain the inner structural bondline temperature below critical material limits (typically $< 175^\circ\text{C}$ for aluminum or composite structures) while enduring surface temperatures that can exceed 2000°C (Bouslog, 2023).

Unlike other subsystems where redundancy can be engineered (e.g., avionics or propulsion), the TPS is often a "single point of failure" system; any breach or significant failure in the thermal barrier typically results in the loss of the vehicle (Anderson, 2006).

TPS technologies are broadly categorized based on their mechanism of heat management:

1. **Ablative TPS:** Designed for the most severe heating environments (e.g., steep re-entry or planetary probes). These materials manage heat loads through phase change (pyrolysis), chemically decomposing to create a char layer and releasing gases that thicken the boundary layer and block convective heating (blowing effect). Examples include PICA (Phenolic Impregnated Carbon Ablator) and Avcoat (Bouslog, 2023).
2. **Reusable TPS:** Designed for missions requiring multiple flights with minimal refurbishment (e.g., Space Shuttle, X-37B). These systems rely on high-emissivity coatings to re-radiate a significant portion of the incident heat flux back into the atmosphere ($q_{rad} \propto \epsilon T^4$) while acting as an insulator to retard heat conduction to the substructure. They do not undergo phase change and are typically limited to lower heat fluxes than ablators.

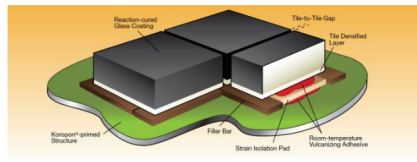
Historically, TPS has been synonymous with "rigid" architectures—tiles, bricks, or heavy monolithic shields. However, the geometric limitations of rigid aeroshells (constrained by the launch vehicle fairing diameter) have necessitated the development of flexible, deployable architectures like HIADs to enable the landing of heavier payloads on Mars (DiNonno & Cheatwood, 2024).

2.4.1 Conventional Rigid TPS (Ablative and Re-usable tiles)

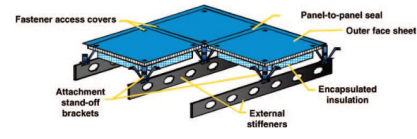
For decades, the standard for atmospheric entry has been rigid TPS, which can be broadly categorized into three architectures based on their functionality and integration with the vehicle structure (Bouslog, 2023).

1. Ablative TPS

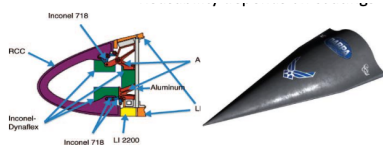
Ablative systems are the workhorses of high-energy planetary entry (e.g., Apollo, Mars Science Laboratory). They function by sacrificing material to manage the ex-



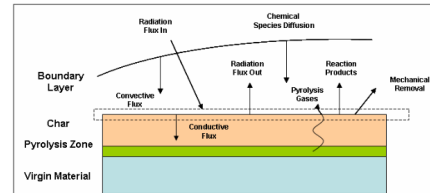
(a): First Type Description



(b): Second Type Description



(c): Third Type Description



(d): Fourth Type Description

Figure 2.6: Classification of Thermal Protection Systems (TPS) Technology (Bouslog, 2023).

treme heat load. As the surface temperature rises, the polymer matrix undergoes pyrolysis—an endothermic chemical decomposition that absorbs energy. This process generates a porous char layer that radiates heat away ($\epsilon \approx 0.9$) and releases pyrolysis gases (blowing) that thicken the boundary layer, physically pushing the high-temperature shock layer away from the wall.

- **Example:** PICA (Phenolic Impregnated Carbon Ablator) used on SpaceX Dragon and NASA's Stardust mission.
- **Limitation:** They are typically single-use and heavy, and their rigid nature limits the vehicle diameter to the size of the launch fairing.

2. Reusable Surface Insulation (RSI)

Made famous by the Space Shuttle Orbiter, these systems consist of lightweight ceramic tiles or blankets (e.g., LI-900 silica tiles) designed to survive thousands of degrees without ablating. They rely on extremely low thermal conductivity to insulate the aluminum airframe and high surface emissivity to re-radiate up to 90% of the incident heat back into space.

- **Example:** HRSI (High-Temperature Reusable Surface Insulation) black tiles.
- **Limitation:** They are extremely fragile, labor-intensive to install/maintain, and cannot withstand the heat fluxes associated with high-speed lunar or Martian

return trajectories.

3. Hot Structures

In this architecture, the primary load-bearing structure itself serves as the thermal protection. Materials like Carbon-Carbon (C-C) or ceramic matrix composites (CMCs) are used for nose caps and wing leading edges where temperatures are highest ($> 1500^{\circ}\text{C}$).

- **Example:** The Reinforced Carbon-Carbon (RCC) nose cap of the Space Shuttle.

While effective, these rigid technologies fundamentally constrain the ballistic coefficient ($m/C_d A$) of the vehicle. To land heavier payloads on Mars, the drag area (A) must be increased beyond the limits of rigid fairings, necessitating the shift toward flexible, deployable architectures (DiNonno & Cheatwood, 2024).



Figure 2.7: Figure for Conventional Rigid TPS.

2.4.2 Hypersonic Inflatable Aerodynamic Decelerators (HIAD/LOFTID)

To overcome the diameter constraints imposed by launch vehicle fairings, NASA and commercial partners have developed Hypersonic Inflatable Aerodynamic Decelerators (HIADs). A HIAD is a deployable aeroshell consisting of an inflatable structural backbone covered by a Flexible Thermal Protection System (FTPS) (DiNonno & Cheatwood, 2024).

Structural Architecture

The structural core of a HIAD typically comprises a stack of pressurized toroidal tubes (tori) made of high-strength braided synthetic fibers (e.g., Kevlar or Vectran).

These toroids are strapped together to form a blunt cone shape. Unlike a smooth rigid shell, this construction inherently creates a "scalloped" or wavy surface topology, where the radius of each toroid creates a series of peaks (crests) and valleys. This specific geometry is the focus of the aerodynamic analysis in this thesis, as it introduces complex flow interactions not present on smooth bodies.

The LOFTID Mission

The technology reached a major milestone with the Low-Earth Orbit Flight Test of an Inflatable Decelerator (LOFTID), successfully conducted in August 2015. The LOFTID vehicle, a 6-meter diameter aeroshell, demonstrated the ability of an inflatable structure to survive the hypersonic heat pulse of re-entry ($> 1400^{\circ}\text{C}$) and successfully decelerate a payload from orbital velocity (DiNonno & Cheatwood, 2024).

LOFTID validated two critical advantages of this technology:

- **Packaging Efficiency:** The entire 6-meter aeroshell was packed into a space significantly smaller than its deployed diameter, allowing it to fit within a standard Atlas V payload fairing.
- **Ballistic Coefficient Management:** By deploying a large drag area (A), the ballistic coefficient ($\beta = m/C_d A$) is drastically reduced. This allows deceleration to occur higher in the atmosphere where density (and thus heating) is lower, a capability essential for landing heavy payloads on the high-elevation surface of Mars.

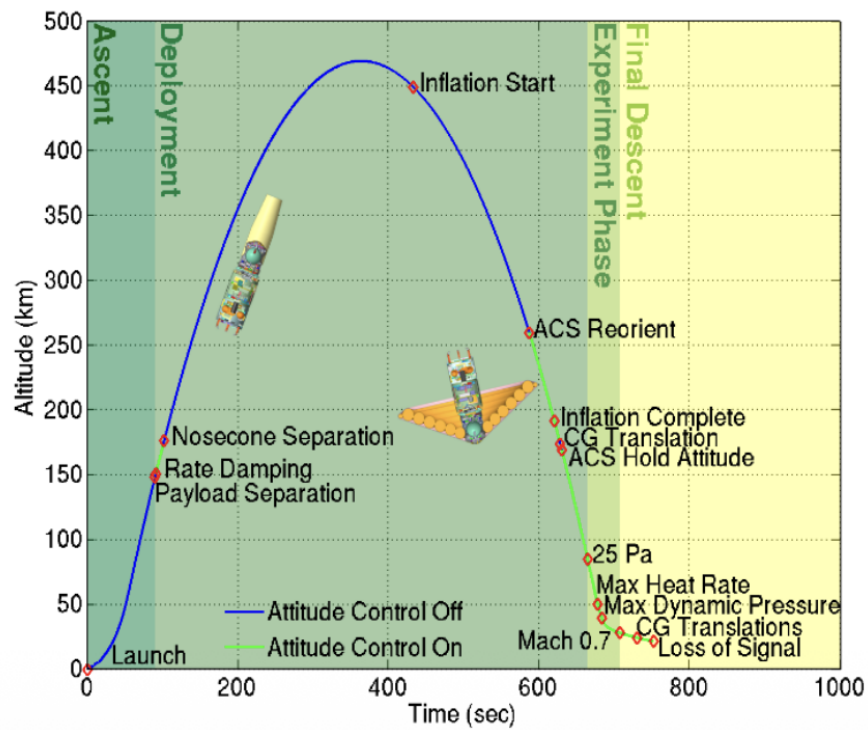


Figure 2.8: LOFTID Mission Trajectory



Figure 2.9: Figure for HIAD/LOFTID.

2.4.3 Comparison of Rigid vs. Flexible Structures

Link with TPS AIAA or Nasa definition of TPS State of the Industry 3 types Jump from TPS-StateofInd-SAMPE-Final2 -> to 3 types example and it's effectiveness Paper

The transition from conventional rigid aeroshells to flexible inflatable decelerators represents a paradigm shift in entry system design. While rigid TPS relies on a fixed outer mold line (OML) to maintain aerodynamic stability, flexible structures introduce a dynamic coupling between the aerodynamic loads and the structural response (DiNonno & Cheatwood, 2024).

The primary distinction lies in the management of the Ballistic Coefficient (β). Rigid vehicles are volume-constrained by the launch shroud, forcing a high β which results in peak heating occurring deep in the atmosphere. In contrast, HIADs decouple the entry diameter from the launch diameter, allowing for a much lower β . This enables deceleration at higher altitudes, significantly reducing the heat flux requirements.

However, the flexibility of the IAD introduces complexities absent in rigid designs. The "scalloped" surface of the stacked toroids creates a rough aerodynamic profile that can augment local heating, and the structural deformation under drag loads can alter the vehicle's stability characteristics.

A comparative summary of the two architectures is presented in Table 2.2.

Table 2.2: Comparison of Conventional Rigid TPS and Hypersonic Inflatable Aerodynamic Decelerators (HIAD).

Parameter	Rigid Aeroshell (Capsule)	Flexible/Inflatable (HIAD)
Launch Constraint	Diameter limited by launch vehicle fairing (e.g., 4-5 m).	Packable; deployed diameter can exceed fairing size (e.g., > 10 m).
Surface Topology	Smooth, continuous Outer Mold Line (OML).	Scalloped/Wavy surface due to stacked toroids.
Aerodynamic Drag	Lower drag area (A); higher Ballistic Coefficient.	Massive drag area (A); lower Ballistic Coefficient.
Thermal Management	Ablation (phase change) or Re-radiation (tiles).	Flexible insulation layers (FTPS) with gas barriers.
Structural Response	Static; shape remains constant under load.	Dynamic; structure deforms/deflects under aerodynamic load (FSI).
Mission Profile	Lunar return, LEO return, small Mars probes.	Heavy-mass Mars landers, engine recovery.

2.5 Impact of Surface Topology on Flow Physics

Topology of Aerodynamics Scalloped physics (Sidenote: Can we compare this with a golf ball theorem then we can derived it from there)

The aerodynamic design of conventional re-entry vehicles typically prioritizes smooth, continuous Outer Mold Lines (OML) to ensure predictable boundary layer development and uniform heat distribution. However, the structural architecture of a HIAD, composed of stacked pressurized toroids, inherently introduces a "scalped" or wavy surface topology.

This deviation from a smooth surface fundamentally alters the flow physics. To understand these effects, it is useful to draw a comparison with the classic "golf ball" phenomenon in fluid mechanics. In the subsonic regime, the dimples on a golf ball act as surface roughness elements that trip the laminar boundary layer into turbulence. This energized turbulent boundary layer is better able to resist adverse pressure gradients, thereby delaying flow separation and significantly reducing pressure drag (form drag).

However, in the hypersonic regime of an IAD, the physics of surface waviness serve a completely different dynamic:

- **Scale:** The "valleys" between toroids are macro-scale features, not micro-roughness.
- **Objective:** Unlike a golf ball where the goal is drag *reduction*, the goal of an IAD is drag *maximization*.
- **Thermal Consequence:** While turbulence on a golf ball is purely beneficial for aerodynamics, turbulence on a hypersonic vehicle creates a severe penalty: it can increase surface heat flux by a factor of 3 to 5 compared to laminar flow (Anderson, 2006).

Therefore, the study of the scalped topology is not focused on drag reduction, but rather on the **Aerothermal Augmentation**—specifically, how the peaks (crests) and valleys of the toroids modify the shock structure and local heating rates. (White, 2006)

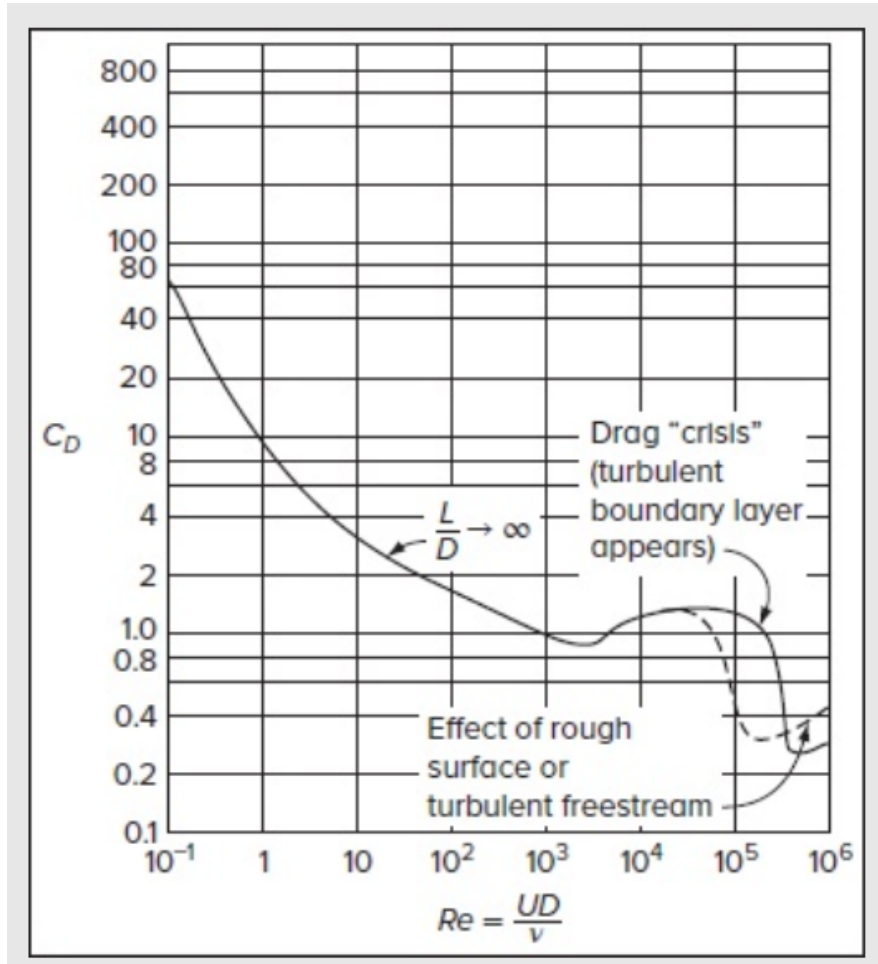


Figure 2.10: Figure for Impact of Surface Topology on Re Number.

2.5.1 Flow over Wavy and Scalloped Walls

The surface of an IAD can be mathematically approximated as a periodic wavy wall or a series of cavity steps. As the hypersonic flow traverses this undulating surface, it encounters alternating regions of favorable and adverse pressure gradients, a phenomenon fundamentally distinct from flow over a flat plate. The physics can be derived by analyzing the momentum equation along the streamline. As the fluid moves over a convex crest (the peak of a toroid), the flow area effectively constricts, causing the flow to accelerate. According to Bernoulli's principle (and its compressible counterpart), this acceleration results in a local decrease in static pressure:

$$\frac{dp}{dx} < 0 \quad (\text{Favorable Gradient - Crest}) \quad (2.13)$$

Conversely, as the flow descends into the concave valley between toroids, the flow area expands and the fluid decelerates. This deceleration generates a localized high-pressure region:

$$\frac{dp}{dx} > 0 \quad (\text{Adverse Gradient - Valley}) \quad (2.14)$$

This creates a recurring cycle of expansion and compression.

- **Over the Crests:** The flow undergoes an expansion fan process, turning away from the shock. The boundary layer thins, and the shear stress (τ_w) increases, leading to peak heating.
- **In the Valleys:** The adverse pressure gradient forces the boundary layer to thicken. If the gradient is strong enough (i.e., the valley is deep), the momentum of the fluid near the wall is insufficient to overcome the pressure rise. This causes the flow to detach, creating a separation bubble or recirculation zone similar to the flow inside a golf ball dimple, but on a macroscopic scale.

This separation is critical for thermal management. In an ideal scenario, a stable recirculation vortex forms in the valley, allowing the high-energy outer flow to "skim" over the top of the gap. This effectively shields the valley floor from the extreme heat of the freestream, a mechanism analogous to the "trapped vortex" combustor concepts in propulsion.

2.5.2 Shock-Boundary Layer Interaction (SBLI) on Rough Surfaces

FSI Book reference is unknown, require questioning. On a smooth hypersonic vehicle, the boundary layer typically grows monotonically from the stagnation point. However, the scalloped surface of an IAD acts as a "rough" wall with large-scale asperities, forcing a complex coupling between the inviscid shock system and the viscous boundary layer, known as Shock-Wave/Boundary-Layer Interaction (SBLI).

This interaction is driven by the compression waves generated at the windward face of each toroid. As the flow exits a valley and encounters the upstream face of the next toroid, the geometry forces the flow to turn into itself (concave turning). In supersonic flow, this turning generates a compression wave or an oblique shock.

This localized shock wave penetrates the boundary layer, imposing a strong adverse pressure gradient ($\frac{dp}{dx} > 0$).

- **Separation:** If the shock is strong enough, the adverse pressure gradient can propagate upstream through the subsonic portion of the boundary layer, causing the flow to separate before it even reaches the valley floor.
- **Reattachment Shock:** When the separated shear layer reattaches to the crest of the next toroid, the flow is forced to align with the wall again. This re-alignment generates a Reattachment Shock.

The intersection of this reattachment shock with the boundary layer is a critical design concern. At the point of reattachment, the boundary layer is extremely thin (high velocity gradient), and the impinging shock further compresses the fluid. This combination results in a localized spike in both pressure and heat flux, which can be significantly higher than the stagnation point heating values predicted for a smooth body (Anderson, 2006).(Wang et al., 2023)

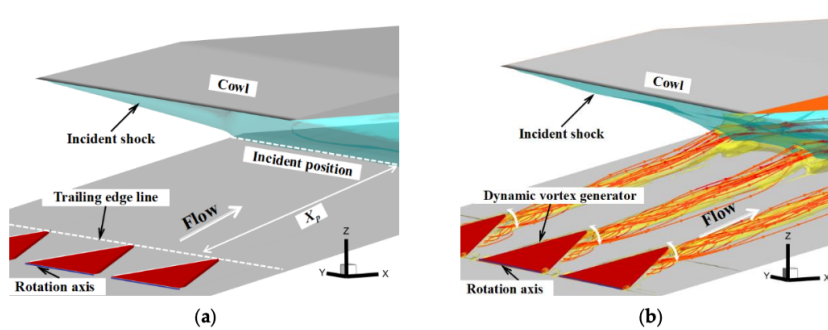


Figure 2.11: Figure for Shock-Boundary Layer Interaction (SBLI) or (SWBLI) on Rough Surfaces.

2.5.3 Recirculation Zones and Stagnation in Surface Valleys

Recirculation Book reference is unknown, require questioning.

The Shear Layer and Vortex Formation

This free shear layer acts as a fluid dynamic boundary.

- **Shear Layer:** A thin region of high vorticity where the velocity transitions from the freestream speed (u_e) to the near-zero velocity of the cavity.
- **Primary Vortex:** Driven by the shear stress at the interface, the fluid mass trapped within the valley is set into rotational motion, creating a large, stable recirculation structure known as the primary vortex.

Stagnation and Heat Alleviation

Ideally, this recirculation zone creates a "dead air" region. The fluid deep within the valley has very low kinetic energy compared to the freestream. Consequently, the convective heat transfer is significantly reduced because the shear layer shields the valley floor from the high-enthalpy gradients of the shock layer.

However, the stability of this zone is dependent on the geometry of the valley, specifically the length-to-depth ratio (L/D).

- **Open Cavity ($L/D < 10$):** The shear layer bridges the entire gap and reattaches at the downstream corner (next crest). This is the desired state for an IAD, as it maintains low heating in the valley.
- **Closed Cavity ($L/D > 10$):** If the valley is too wide or shallow, the shear layer may curve inward and impinges on the valley floor. This stagnation of the shear layer on the bottom surface would result in a local "hot spot" of high pressure and heating, negating the thermal benefits of the scalloped design.

This is as for example what is recirculation effect on an structure (Petronio, Roman, Nasello, & Armenio, 2013)

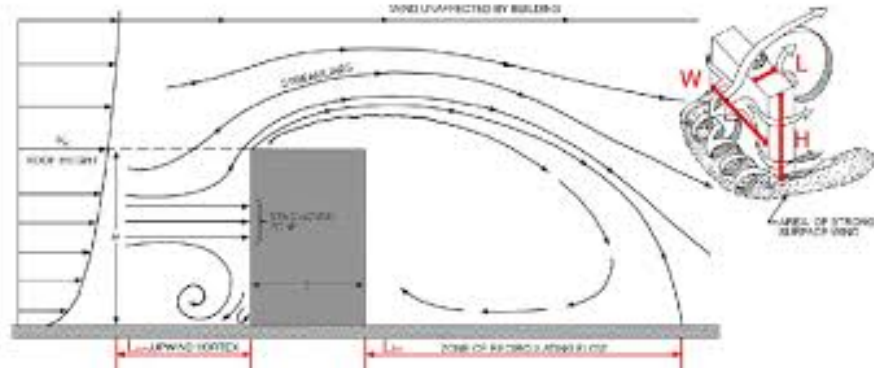


Figure 2.12: Figure for Recirculation Zones and Stagnation Valleys Example on different scenarios.

2.5.4 Localized Peak Heating Effects

Link with Hypersonic and High Temperature Gas Book (3 Book Linkage) (derivation on the mathematics of stagnation point of Mach number) While the recirculation zones in the valleys may alleviate thermal loads, the protruding crests of the toroidal rings experience the opposite effect: severe localized heating. This phenomenon is governed by the mathematics of stagnation point flow, where the local geometry creates conditions significantly more severe than the average heating over the vehicle.

The Stagnation Temperature (Energy Potential)

The potential for heat transfer is driven by the total energy of the flow. As the fluid decelerates to zero velocity at a stagnation point (such as the windward face of a toroid), its kinetic energy is converted to enthalpy. For a calorically perfect gas, the stagnation temperature T_0 is a function of the freestream Mach number M_∞ (Anderson, 2006):

$$\frac{T_0}{T_\infty} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \quad (2.15)$$

For a hypersonic flow of $Ma = 5$ (where $\gamma = 1.4$), the stagnation temperature is six times the freestream temperature. This T_0 represents the driving potential for heat transfer.

The Geometric Scaling (Heat Flux Rate)

While T_0 determines the *temperature* limit, the *rate* at which heat enters the structure (q_s) is determined by the velocity gradient at the wall. According to the classical Fay-Riddell correlation, the stagnation point heat flux scales inversely with the square root of the nose radius (R_n):

$$q_s \approx 0.76 \cdot Pr^{-0.6} \cdot (h_0 - h_w) \cdot \sqrt{\rho_e \mu_e} \cdot \sqrt{\left(\frac{du_e}{dx}\right)_s} \quad (2.16)$$

The velocity gradient term $\left(\frac{du_e}{dx}\right)_s$ for a blunt body is geometrically dependent on the radius of curvature R :

$$\left(\frac{du_e}{dx}\right)_s \approx \frac{V_\infty}{R} \sqrt{\frac{2\rho_\infty}{\rho_0}} \quad (2.17)$$

Substituting this back into the heat flux equation yields the critical proportionality:

$$q_s \propto \sqrt{\frac{1}{R}} \quad (2.18)$$

This derivation highlights the critical vulnerability of scalloped IAD geometries. While the *overall* effective radius of the vehicle is large (low global heating), the *local* radius of an individual toroid tube (R_{tube}) is small. Consequently, the crest of each toroid acts as a local stagnation point with a small radius, leading to "peak heating" spikes that can be significantly higher than the heating on a smooth sphere-cone of the same size.

It is demonstrated on the analogue of the localized heating (Daub, Willems, & Gülhan, 2022)

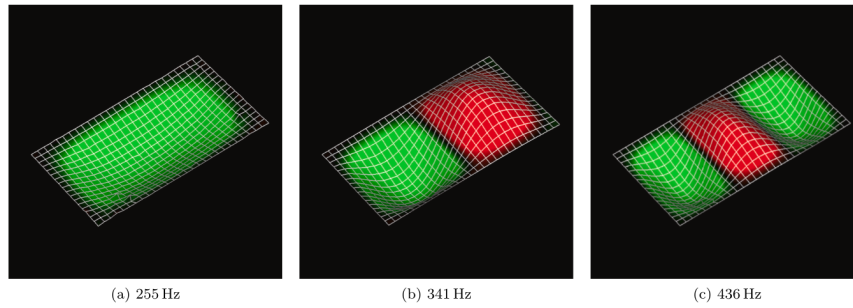


Figure 2.13: Figure for Localized Peak Heating Effects in Amplification but through Oscillation.

2.6 Theoretical Framework of CFD

Computational Fluid Dynamics (CFD) serves as the third pillar of fluid dynamics research, complementing theoretical analysis and experimental testing. It involves the systematic application of numerical methods to solve the governing equations of fluid motion, transforming the continuous mathematical models into discrete algebraic equations that can be solved computationally (Versteeg & Malalasekera, 2007).

For hypersonic research, where flight tests are prohibitively expensive and ground-based facilities (such as wind tunnels) often suffer from scaling limitations or wall interference, CFD provides a critical capability to resolve complex flow physics. By discretizing the spatial domain into a grid, CFD allows for the detailed interrogation of flow properties—such as shock structures and thermal gradients—that are often difficult to measure experimentally.

The theoretical foundation of this analysis rests upon the Continuum Hypothesis. This assumption treats the fluid not as a collection of discrete molecules, but as a continuous substance where macroscopic properties (density, pressure, velocity, and temperature) are well-defined at every point in space and time. Based on this hypothesis, the flow physics are governed by three fundamental conservation laws:

- Conservation of Mass (Continuity)
- Conservation of Momentum (Newton’s Second Law)
- Conservation of Energy (First Law of Thermodynamics)

2.6.1 The Navier-Stokes Equations (Conservation of Momentum)

The conservation of momentum is based on Newton's Second Law of Motion, which states that the rate of change of momentum of a fluid particle is equal to the sum of the forces acting on it.

For an infinitesimal fluid element of volume $dV = dxdydz$, the forces are categorized into two types:

- **Surface Forces:** Forces acting on the boundaries of the element, including pressure (p) and viscous stresses (τ).
- **Body Forces:** Forces acting on the entire mass of the element (e.g., gravity), denoted by the source term S_M .

1. The Momentum Balance

Consider the x -component of momentum. The net force due to surface stresses in the x -direction is the sum of the gradients of the normal stress (σ_{xx}) and shear stresses (τ_{yx}, τ_{zx}). The general differential form for the conservation of momentum in the i -direction (where $i = x, y, z$) is given by (Versteeg & Malalasekera, 2007):

$$\frac{\partial(\rho u_i)}{\partial t} + \nabla \cdot (\rho u_i \mathbf{u}) = \nabla \cdot \sigma_{ij} + S_{M,i} \quad (2.19)$$

Where the left-hand side represents the rate of change of momentum (transient + convective), and the right-hand side represents the forces. The stress tensor σ_{ij} includes the thermodynamic pressure and the viscous stress tensor τ_{ij} :

$$\sigma_{ij} = -p\delta_{ij} + \tau_{ij} \quad (2.20)$$

2. Constitutive Relations (Newtonian Fluid)

To close the system of equations, the viscous stresses must be related to the velocity gradients (strain rates). For a **Newtonian fluid**, the viscous stress is proportional to the rate of deformation. This relationship is defined by:

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \delta_{ij} (\nabla \cdot \mathbf{u}) \quad (2.21)$$

Here, μ is the dynamic viscosity, and λ is the second coefficient of viscosity. Following Stokes' hypothesis for monoatomic gases, we assume $\lambda = -\frac{2}{3}\mu$. This leads to the final form of the viscous stress tensor for a compressible flow:

$$\tau_{ij} = \mu \left[\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3}(\nabla \cdot \mathbf{u})\delta_{ij} \right] \quad (2.22)$$

3. The Complete Navier-Stokes Equation

Substituting the constitutive relation back into the momentum equation yields the complete Compressible Navier-Stokes Equation. In vector notation, it is written as:

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot \left[\mu \left(\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right) \right] + \mathbf{S}_M \quad (2.23)$$

This equation describes how the velocity field evolves under the influence of pressure gradients, viscous diffusion, and external forces. For hypersonic flows, the compressibility term $(\nabla \cdot \mathbf{u})$ is non-zero and significant, coupling the momentum equation strongly with the energy equation.

2.6.2 Conservation of Mass and Energy Equations

In addition to momentum, the numerical solution must satisfy the fundamental principles of mass conservation and the First Law of Thermodynamics.

1. Conservation of Mass (Continuity Equation)

The continuity equation describes the mass balance within a control volume. It states that the rate of increase of mass inside the fluid element is equal to the net rate of flow of mass into the element across its faces.

For a compressible fluid, where density ρ is a variable, the differential form is given by (Versteeg & Malalasekera, 2007):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (2.24)$$

The first term $\frac{\partial \rho}{\partial t}$ represents the rate of change of density (zero for steady flows), and the second term $\nabla \cdot (\rho \mathbf{u})$ represents the convective mass flux (divergence of the mass flow rate).

2. Conservation of Energy

For hypersonic flows, the energy equation is the most critical component of the governing set, as it couples the flow velocity to the thermodynamic state (temperature). Based on the First Law of Thermodynamics, the rate of change of energy of a fluid particle is equal to the rate of heat added to the fluid plus the rate of work done on the fluid.

The total energy E per unit mass is defined as the sum of internal energy (i) and kinetic energy:

$$E = i + \frac{1}{2}|\mathbf{u}|^2 \quad (2.25)$$

The conservative form of the energy equation is written as (Versteeg & Malalasekera, 2007):

$$\frac{\partial(\rho E)}{\partial t} + \nabla \cdot (\rho E \mathbf{u}) = \nabla \cdot (k \nabla T) + \nabla \cdot (\boldsymbol{\tau}_{ij} \cdot \mathbf{u}) + S_E \quad (2.26)$$

The terms on the right-hand side represent the specific mechanisms of energy transfer:

- Heat Conduction: $\nabla \cdot (k \nabla T)$ represents thermal diffusion governed by Fourier's Law.
- Viscous Dissipation: $\nabla \cdot (\boldsymbol{\tau}_{ij} \cdot \mathbf{u})$ represents the work done by viscous stresses. This term is often negligible in low-speed flows but is the dominant source of aerodynamic heating in hypersonic boundary layers, converting kinetic energy directly into heat.
- Source Terms: S_E accounts for other sources such as chemical reaction heat release or radiation.

2.6.3 Finite Volume Method (FVM) Discretization (The basis of Ansys/Open-FOAM)

The system of partial differential equations (PDEs) derived in the previous sections is analytically unsolvable for arbitrary geometries. To compute a solution, the continuous domain must be discretized. The standard approach employed by modern CFD solvers, including ANSYS Fluent and OpenFOAM, is the Finite Volume Method (FVM) (Versteeg & Malalasekera, 2007).

Unlike Finite Difference methods which discretize the differential form of the equations at grid points, FVM discretizes the integral form of the conservation laws over a set of contiguous control volumes (cells).

The General Transport Equation

The conservation of any scalar property ϕ (such as enthalpy, turbulence kinetic energy, or a velocity component) can be expressed in a generic integral form over a control volume V bounded by a surface S :

$$\underbrace{\frac{\partial}{\partial t} \int_V \rho \phi dV}_{\text{Transient Term}} + \underbrace{\oint_S \rho \phi \mathbf{u} \cdot \mathbf{n} dS}_{\text{Convective Flux}} = \underbrace{\oint_S \Gamma_\phi \nabla \phi \cdot \mathbf{n} dS}_{\text{Diffusive Flux}} + \underbrace{\int_V S_\phi dV}_{\text{Source Term}} \quad (2.27)$$

Where $\mathbf{n}$ is the unit normal vector to the surface, and Γ_ϕ is the diffusion coefficient.

Discretization into Algebraic Equations

The FVM transforms this integral equation into a system of algebraic equations by approximating the surface integrals as the sum of fluxes crossing the faces of the control volume. For a cell P with a set of faces f :

$$\sum_f (\rho \phi \mathbf{u} \cdot \mathbf{n} A)_f = \sum_f (\Gamma_\phi \nabla \phi \cdot \mathbf{n} A)_f + S_\phi V_P \quad (2.28)$$

The solver then solves this linear system $[A]\{x\} = \{b\}$ to find the value of ϕ at the cell centers.

Why FVM is Required for Hypersonics

The primary advantage of FVM for high-speed aerodynamics is the property of conservativeness. The flux leaving a control volume across a specific face is numerically identical to the flux entering the adjacent control volume through that same face. This ensures that mass, momentum, and energy are strictly conserved across the entire domain, even in the presence of flow discontinuities like shock waves, where differential methods often fail (Versteeg & Malalasekera, 2007).

2.7 Turbulence Modeling for High-Speed Flows

Turbulence remains one of the most complex and challenging aspects of fluid dynamics to model accurately. In the context of hypersonic re-entry, the state of the boundary layer—whether laminar, transitional, or fully turbulent—has a profound impact on the aerothermal loads.

While laminar flow is characterized by smooth, ordered fluid motion, turbulent flow involves chaotic, three-dimensional, time-dependent fluctuations. For an Inflatable Aerodynamic Decelerator (IAD), the onset of turbulence is a critical design constraint because turbulent boundary layers possess much higher energy gradients at the wall. Consequently, turbulent heating rates are typically 3 to 5 times higher than laminar heating rates for the same flight condition (Anderson, 2006).

Resolving these fluctuations directly via Direct Numerical Simulation (DNS) requires a grid fine enough to capture the smallest scales of turbulent energy dissipation (the Kolmogorov scales). The computational cost scales with $Re^{9/4}$, making DNS impossible for the high Reynolds number flows ($Re > 10^6$) encountered during re-entry. Therefore, engineering analysis relies on Turbulence Modeling—mathematical approximations that predict the effect of turbulence on the mean flow without resolving the full spectrum of turbulent fluctuations (Versteeg & Malalasekera, 2007).

2.7.1 The Closure Problem (RANS Equations)

To make the calculation of turbulent flows computationally feasible for engineering applications, the standard approach is to solve the Reynolds-Averaged Navier-Stokes (RANS) equations. This method is based on the Reynolds Decomposition, proposed by Osborne Reynolds (1895), which separates the instantaneous flow variables into

a time-averaged mean component and a fluctuating turbulent component (Versteeg & Malalasekera, 2007).

For velocity $\mathbf{u}$ and pressure p :

$$\mathbf{u}(\mathbf{x}, t) = \bar{\mathbf{u}}(\mathbf{x}) + \mathbf{u}'(\mathbf{x}, t) \quad (2.29)$$

$$p(\mathbf{x}, t) = \bar{p}(\mathbf{x}) + p'(\mathbf{x}, t) \quad (2.30)$$

Where $\bar{\mathbf{u}}$ is the mean velocity and $\mathbf{u}'$ is the turbulent fluctuation, with the property that the time average of the fluctuation is zero ($\overline{\mathbf{u}'} = 0$).

Substituting these decomposed variables into the instantaneous Navier-Stokes equations and time-averaging the entire equation yields the RANS Momentum Equation:

$$\frac{\partial(\rho \bar{u}_i)}{\partial t} + \frac{\partial}{\partial x_j}(\rho \bar{u}_i \bar{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \rho \overline{u'_i u'_j} \right] \quad (2.31)$$

While this equation closely resembles the laminar Navier-Stokes equation, it contains a critical new term: $-\rho \overline{u'_i u'_j}$. This term represents the transport of momentum due to turbulent fluctuations and acts as an additional apparent stress on the fluid. It is known as the Reynolds Stress Tensor (τ_{ij}^{Re}).

The Closure Problem

The Reynolds stress tensor is symmetric and introduces six additional unknown components (normal and shear stresses) to the system of equations. However, we do not have six additional equations to solve for them.

- **Unknowns:** Mean pressure $\bar{p}$, three mean velocity components $\bar{u}, \bar{v}, \bar{w}$, and six Reynolds stresses.
- **Equations:** One continuity equation and three momentum equations.

Since the number of unknowns exceeds the number of conservation equations, the system is mathematically "open." The Closure Problem refers to the necessity of mod-

eling these Reynolds stresses as a function of the mean flow variables to close the system and obtain a solution.

2.7.2 Comparison of $k - \varepsilon$ vs. $k - \omega$ models

The most widely used turbulence models in engineering CFD are the two-equation models, which introduce two additional transport equations to compute the eddy viscosity μ_t . The two dominant variants are the Standard $k - \varepsilon$ model and the Wilcox $k - \omega$ model. While both rely on the Boussinesq hypothesis, they differ fundamentally in their formulation of the turbulent length scale (Versteeg & Malalasekera, 2007).

The Standard $k - \varepsilon$ Model

Developed by Jones and Launder, this model solves for turbulent kinetic energy (k) and the rate of dissipation (ε).

- **Strengths:** It is numerically robust, computationally efficient, and well-validated for fully turbulent free-shear flows (e.g., wakes and jets).
- **Weaknesses:** Its formulation is invalid in the near-wall region where viscous effects dominate. To bridge the gap between the turbulent core and the wall, it relies on empirical "wall functions," which degrades accuracy for flows with separation or severe pressure gradients. Consequently, it often over-predicts heat flux and delays separation in adverse pressure gradients (Versteeg & Malalasekera, 2007).

The Wilcox $k - \omega$ Model

Developed by Wilcox, this model solves for k and the specific dissipation rate (ω).

- **Strengths:** The $k - \omega$ formulation is mathematically stable in the viscous sub-layer and can be integrated directly to the wall without damping functions. This allows for superior prediction of boundary layer separation and wall heat transfer, making it ideal for the scalloped valleys of an IAD.
- **Weaknesses:** It suffers from a critical deficiency known as "freestream sensitivity." The results inside the boundary layer depend strongly on the arbitrary value of ω specified at the far-field inlet. For external aerodynamics, this can lead to unacceptable errors in drag and lift prediction.

For the analysis of a HIAD, neither model is sufficient on its own. The $k - \varepsilon$ model cannot accurately resolve the complex recirculation in the valleys, while the standard $k - \omega$ model is too unreliable for the external shock layer. This limitation necessitates the use of a hybrid formulation. (Saleem et al., 2023)

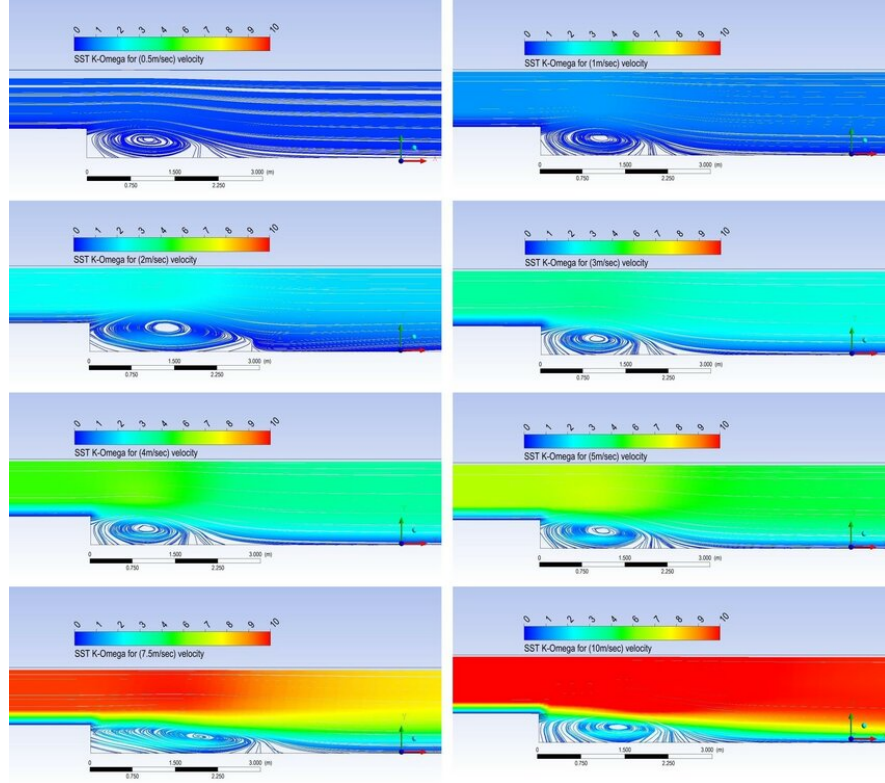


Figure 2.14: Figure for Comparison of k-epsilon vs. k-omega models.

2.7.3 The SST (Shear Stress Transport) $k - \omega$ Model (Why it is chosen for Wall Heat Flux)

Paper

To overcome the limitations of the standard two-equation models, Menter (1994) developed the Shear Stress Transport (SST) $k - \omega$ model. This model effectively blends the robust and accurate formulation of the $k - \omega$ model in the near-wall region with the freestream independence of the $k - \varepsilon$ model in the far field.

1. Zonal Blending Functions

The core innovation of the SST model is the use of a blending function, F_1 , which transitions the model formulation based on the distance from the wall (y).

- **Near Wall** ($F_1 = 1$): The model activates the standard $k - \omega$ equations, allowing for direct resolution of the viscous sublayer without damping functions. This ensures accurate prediction of wall heat flux (q_w) and wall shear stress (τ_w).
- **Far Field** ($F_1 = 0$): The model smoothly transitions to the $k - \varepsilon$ formulation (mathematically transformed into $k - \omega$ variables). This eliminates the sensitivity to freestream turbulence properties that plagues the standard $k - \omega$ model.

2. The Shear Stress Limiter

In flows with strong adverse pressure gradients—such as the deceleration zone in the valleys of a scalloped IAD—standard two-equation models often over-predict the eddy viscosity (μ_t). This leads to a failure to predict flow separation, resulting in erroneous aerodynamic coefficients.

The SST formulation introduces a limiter to the eddy viscosity definition to satisfy the Bradshaw assumption (that shear stress is proportional to kinetic energy in boundary layers):

$$\mu_t = \frac{\rho k}{\omega} \frac{1}{\max\left(\frac{1}{\alpha^*}, \frac{SF_2}{a_1 \omega}\right)} \quad (2.32)$$

Where S is the strain rate magnitude and F_2 is a second blending function. This limiter prevents the buildup of excessive turbulence levels in stagnation regions and separation zones.

Justification for HIAD Analysis

For this research, the SST $k - \omega$ model is selected as the turbulence closure for the following reasons:

1. **Heat Transfer Accuracy:** It is widely regarded as the industry standard for aerodynamic heating predictions (Menter, 1994), typically predicting heat flux within

5-10% of experimental data for attached flows.

2. **Separation Prediction:** The shear stress transport formulation is critical for accurately capturing the size and stability of the recirculation vortices inside the toroidal valleys, which directly drive the local thermal environment.

2.8 Thermodynamic Gas Models

In standard aerodynamic analysis (subsonic or supersonic), air is typically modeled as a Calorically Perfect Gas. This assumes that the specific heats (C_p and C_v) are constant and that the internal energy is a linear function of temperature ($e = C_v T$). However, for a vehicle entering the atmosphere at hypersonic speeds ($Ma > 5$), this assumption breaks down catastrophically (Anderson, 2006).

The immense kinetic energy converted into heat across the bow shock raises the gas temperature to levels where the internal structure of the air molecules is altered. As the temperature rises, the gas undergoes a sequence of energy mode excitations:

- $T > 800 \text{ K}$: Vibrational excitation of N_2 and O_2 molecules becomes significant. The gas becomes Thermally Perfect but Calorically Imperfect ($C_p = f(T)$).
- $T > 2000 \text{ K}$: Oxygen molecules begin to dissociate ($O_2 \rightarrow 2O$).
- $T > 4000 \text{ K}$: Nitrogen molecules begin to dissociate ($N_2 \rightarrow 2N$).
- $T > 9000 \text{ K}$: Atoms begin to ionize ($N \rightarrow N^+ + e^-$).

Accurately modeling these phenomena is essential for an Inflatable Aerodynamic Decelerator (IAD). Failure to account for variable specific heat, for instance, can lead to an over-prediction of the post-shock temperature and a significant under-prediction of the density, resulting in incorrect shock stand-off distances and non-physical heat flux results.

2.8.1 The Equation of State: Ideal Gas vs. Real Gas Effects

The Equation of State relates the thermodynamic properties of pressure (p), density (ρ), and temperature (T). In standard aerodynamic analysis, the air is treated as an Ideal Gas, obeying the relation:

$$p = \rho RT \quad (2.33)$$

Where R is the specific gas constant ($R = 287 \text{ J/kg}\cdot\text{K}$ for air). This relation assumes that intermolecular forces are negligible and the volume of the molecules themselves is zero (Anderson, 2006).

In the context of hypersonic re-entry, the term "Real Gas Effects" is often used broadly to describe deviations from standard air behavior. However, it is important to distinguish between two types of deviations:

1. **Dense Gas Effects (Intermolecular Forces):** At extremely high pressures (e.g., deep atmospheric entry on Gas Giants), gas molecules are packed so tightly that intermolecular forces become significant. In this regime, the Ideal Gas Law is invalid, and a compressibility factor Z is introduced ($p = \rho ZRT$). For Earth re-entry at high altitudes (low density), these effects are typically negligible, and the Ideal Gas EOS ($p = \rho RT$) remains valid.
2. **High-Temperature Effects (Chemistry):** This is the dominant "Real Gas" effect for HIADs. As temperature increases, the chemical composition of the air changes (dissociation). While the equation $p = \rho R_{mix} T$ still holds, the gas constant R_{mix} is no longer constant because the molecular weight of the mixture changes as N_2 and O_2 break apart.

A comparison of these modeling assumptions is presented in Table 2.3.

Table 2.3: Comparison of Gas Models in Hypersonic Aerodynamics.

Feature	Ideal (Perfect) Gas Model	Real Gas (Hypersonic) Model
Equation of State	$p = \rho RT$ (Constant R)	$p = \rho R_{mix} T$ (Variable R due to chemistry)
Specific Heats	Constant ($C_p, C_v = \text{const}$)	Variable ($C_p(T)$) due to vibrational excitation.
Internal Energy	$e = C_v T$ (Linear)	$e = f(T)$ (Non-linear, includes chemical potential).
Shock Density Ratio	Limited to $(\gamma + 1)/(\gamma - 1) = 6$	Can exceed 10 or 12 (Thinner shock layers).
Temperature Prediction	Over-predicts post-shock temperature.	Lower temperature (Energy absorbed by chemistry).

2.8.2 Calorically Perfect vs. Thermally Perfect Gas (Variable C_p)

The distinction between a calorically perfect and a thermally perfect gas lies in the behavior of the specific heat capacities (C_p and C_v) as the temperature of the fluid increases.

1. Calorically Perfect Gas (CPG)

At low temperatures (typically $T < 800$ K), the vibrational energy modes of the nitrogen and oxygen molecules are "frozen." The internal energy of the gas consists only of translational and rotational kinetic energy, which are fully excited at room temperature. Consequently, the specific heat at constant pressure (C_p) is constant, and the internal energy (e) is a linear function of temperature:

$$e = C_v T \quad \text{and} \quad h = C_p T \quad (2.34)$$

This assumption is valid for subsonic and low-supersonic flows but leads to significant errors in the hypersonic regime.

2. Thermally Perfect Gas (TPG)

As the gas temperature rises across a strong bow shock ($T > 800$ K), the collisions between molecules become energetic enough to activate the vibrational modes of the

diatomic molecules (N_2 and O_2). These vibrational modes absorb a portion of the flow's kinetic energy, acting as an additional "heat sink."

Mathematically, this means that the specific heats become functions of temperature:

$$C_p = f(T) \quad \text{and} \quad C_v = f(T) \quad (2.35)$$

However, the gas still obeys the thermal equation of state $p = \rho RT$ (assuming no chemical dissociation). A gas that satisfies $p = \rho RT$ but has variable specific heats is defined as a Thermally Perfect Gas.

In CFD simulations for hypersonic re-entry, $C_p(T)$ is typically modeled using a polynomial curve fit, such as the NASA 7-coefficient or 9-coefficient polynomials:

$$\frac{C_p(T)}{R} = a_1 + a_2T + a_3T^2 + a_4T^3 + a_5T^4 \quad (2.36)$$

Using a Thermally Perfect Gas model is essential for the HIAD analysis because the absorption of energy into vibrational modes lowers the peak shock layer temperature compared to a Calorically Perfect prediction, resulting in more accurate (and often less conservative) density and heat flux calculations (Anderson, 2006).

2.8.3 Temperature-Dependent Viscosity (Sutherland's Law)

In classical low-speed aerodynamics, fluid viscosity is often treated as a constant property. However, in the hypersonic regime where temperature variations within the shock layer can span thousands of degrees, the viscosity of the gas changes largely. Unlike liquids, where viscosity decreases with heat, the viscosity of a gas *increases* with temperature due to the increased molecular agitation and momentum exchange between fluid layers.

To accurately model the viscous stresses and heat transfer in the boundary layer, the dynamic viscosity (μ) must be coupled to the local static temperature (T). The standard model used in compressible flow CFD is Sutherland's Law (1893), which is derived from the kinetic theory of gases using an idealized intermolecular potential (Anderson, 2006).

$$\mu(T) = \mu_{ref} \left(\frac{T}{T_{ref}} \right)^{3/2} \frac{T_{ref} + S}{T + S} \quad (2.37)$$

Where:

- μ is the dynamic viscosity at temperature T .
- μ_{ref} is the reference viscosity at reference temperature T_{ref} .
- S is the Sutherland constant (effective temperature), which accounts for inter-molecular attraction.

For standard air, the coefficients used in this study are:

- $\mu_{ref} = 1.716 \times 10^{-5} \text{ kg/(m}\cdot\text{s)}$
- $T_{ref} = 273.15 \text{ K}$
- $S = 110.4 \text{ K}$

Implementing Sutherland's Law is vital for the thermal analysis of an IAD. Since aerodynamic heating is directly proportional to the square root of the viscosity product ($q_w \propto \sqrt{\rho\mu}$), neglecting the temperature dependence of μ would result in a gross underestimation of the surface heat flux.

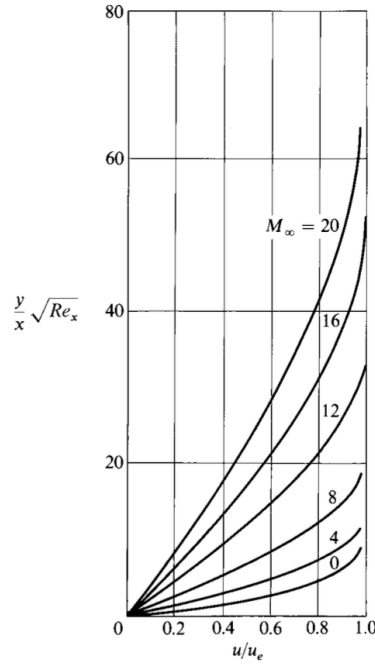


Figure 2.15: (Sutherland's Law effect on Laminar Boundary Layer Insulated Flat Plate).

2.9 Numerical Solvers and Schemes

The numerical simulation of hypersonic flow fields requires a solver architecture capable of handling the strong coupling between the velocity field, density, and energy equation. For this study, the Density-Based Coupled Solver is selected over the traditional Pressure-Based solver (Versteeg & Malalasekera, 2007).

2.9.1 Density-Based Solver Formulation

In low-speed (incompressible) flows, density is constant, and the pressure field is derived from the continuity equation (typically using algorithms like SIMPLE or PISO). However, in hypersonic flows ($Ma > 5$), density is a variable function of pressure and temperature (via the Equation of State).

The Density-Based solver integrates the continuity, momentum, energy, and species transport equations simultaneously (coupled) rather than sequentially.

- **Coupling:** The governing equations are solved as a vector system. This robust coupling is essential for resolving the high-gradient shock structures where den-

sity changes abruptly.

- **Time Formulation:** An Implicit formulation is employed to allow for larger Courant-Friedrichs-Lewy (CFL) numbers while maintaining unconditional stability during the convergence process towards a steady-state solution.

2.9.2 Flux Discretization Schemes

To compute the fluxes of mass, momentum, and energy across the cell faces, specific discretization schemes are chosen to balance numerical stability with accuracy.

1. Convective Flux Type: Roe-FDS

The Roe Flux-Difference Splitting (Roe-FDS) scheme is selected for the inviscid flux formulation. The Roe scheme is a Godunov-type upwind method that solves an approximate Riemann problem at each cell face.

- **Justification:** It is widely renowned for its superior shock-capturing capabilities, providing crisp resolution of discontinuities (shocks) without introducing excessive numerical diffusion (smearing) (Versteeg & Malalasekera, 2007).

2. Spatial Discretization: Second-Order Upwind

To ensure the solution is grid-independent and physically accurate:

- **Gradients:** The Least Squares Cell-Based method is used to compute gradients, as it is less sensitive to grid skewness on the unstructured mesh of the scalloped toroids.
- **Flow Variables:** A Second-Order Upwind scheme is applied to the flow, turbulence, and energy equations. This scheme reconstructs face values using a Taylor series expansion from the cell center, offering significantly higher accuracy than First-Order schemes, which are prone to false diffusion that can artificially thicken the bow shock.

2.9.3 Shock Capturing Schemes (Upwind, Flux splitting)

The defining characteristic of hypersonic flow is the presence of strong discontinuities (shock waves) where flow properties change abruptly over a microscopic dis-

tance. Standard central-difference schemes, commonly used in subsonic flow, are non-dissipative and essentially "blind" to the direction of flow propagation. When applied to a shock wave, they invariably produce unphysical oscillations (Gibbs phenomenon) that lead to solution divergence.

To resolve these discontinuities without numerical instability, modern CFD relies on Shock Capturing techniques rooted in the mathematical theory of hyperbolic conservation laws (Versteeg & Malalasekera, 2007).

The Upwind Philosophy

The physical basis for upwinding is the transport of information. In supersonic flow, disturbances cannot propagate upstream against the flow; information travels only along characteristic lines from upstream to downstream.

- **Concept:** The numerical flux at a cell face should be calculated using the state variables from the "upwind" side (the side from which the fluid is flowing).
- **Implementation:** First-order upwind schemes simply take the cell center value of the upstream cell. While stable, this is highly diffusive, causing sharp shock waves to appear "smeared" across many grid cells.
- **High-Resolution Schemes:** To achieve accuracy, Second-Order Upwind schemes are employed. These reconstruct the face value using a Taylor series expansion based on upstream gradients. To prevent oscillations near shocks, Flux Limiters (e.g., Min-Mod, Barth-Jespersen) are applied to strictly enforce monotonicity (TVD - Total Variation Diminishing properties).

Flux Splitting Schemes

To mathematically determine "which way is upstream" for the complex system of Navier-Stokes equations (where sound waves can travel in multiple directions), Flux Splitting methods are used. These schemes decompose the total flux into wave components.

1. Flux Vector Splitting (FVS): Methods like *Steger-Warming* or *Van Leer* split the flux vector into positive (forward-moving) and negative (backward-moving) com-

ponents based on the sign of the eigenvalues (wave speeds).

$$F = F^+(\lambda^+) + F^-(\lambda^-) \quad (2.38)$$

While robust, FVS schemes are known to be excessively dissipative in boundary layers, often over-predicting skin friction and heat transfer.

2. Flux Difference Splitting (FDS) - The Preferred Choice: Methods like Roe's Scheme (Approximate Riemann Solver) model the interface between two cells as a localized shock tube problem (Riemann problem).

- Instead of splitting the flux itself, Roe's scheme solves for the difference in flux across the wave structure.
- **Relevance to HIADs:** FDS schemes are significantly less dissipative than FVS schemes. This makes them the standard choice for hypersonic viscous flows because they can accurately capture the crispness of the bow shock while simultaneously resolving the delicate shear gradients in the boundary layer (Toro, 2009).

3. AUSM (Advection Upstream Splitting Method): A hybrid approach often used in modern codes (like ANSYS Fluent's "AUSM+" option). It splits the flux into a convective term (treated as a scalar) and a pressure term (treated acoustically). This method is designed to be as accurate as Roe's scheme but more robust against the "carbuncle phenomenon" (a numerical instability often seen near the stagnation point of blunt bodies).

CHAPTER 3

METHODOLOGY

3.1 Research Design and Workflow

This chapter outlines the computational framework utilized to investigate the aerothermal effects of surface topology on a Hypersonic Inflatable Aerodynamic Decelerator (HIAD). The research workflow follows a sequential numerical analysis approach, moving from parametric geometry generation to high-fidelity Computational Fluid Dynamics (CFD) simulation.

To systematically isolate the effects of the "scalloped" surface geometry, the methodology is designed to compare a baseline rigid sphere-cone against the proposed inflatable architecture under identical hypersonic flight conditions. The workflow is visualized in Figure 3.1 and consists of four primary stages:

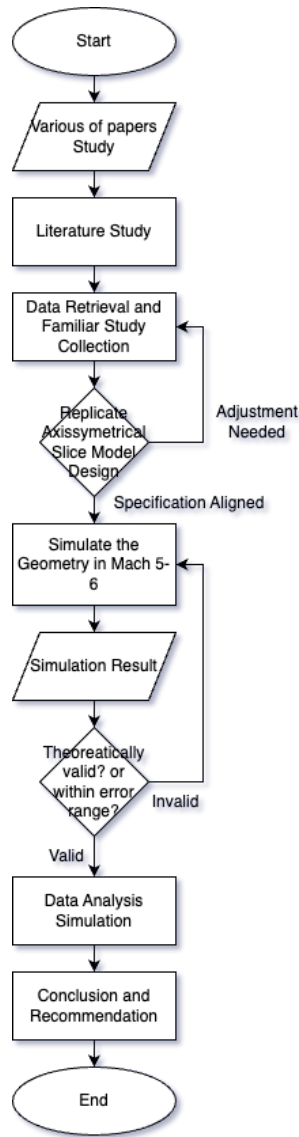


Figure 3.1: Flowchart of the numerical research methodology.

The process is divided into four primary stages:

1. **Geometric Modeling:** Automated generation of the scalloped (inflatable) and smooth (baseline) profiles using parametric Python scripting in FreeCAD to ensure geometric consistency.
2. **Domain Discretization:** Generation of a structured/hybrid mesh with strict near-wall refinement ($y^+ \approx 1$) to resolve the high thermal gradients in the boundary layer.

3. **Numerical Configuration:** Setup of the density-based coupled solver, utilizing the SST $k - \omega$ turbulence model and temperature-dependent gas properties (Sutherland's Law).
4. **Validation and Analysis:** Verification of the solution through a grid independence study and validation of the thermal results against theoretical Fay-Riddell correlations.

3.2 Geometric Modeling and Computational Domain

3.2.1 Parametric Geometry Generation

To ensure geometric consistency, the vehicle profiles were generated using a parametric Python script within the FreeCAD environment. The geometric constraints are derived from the NASA IRVE-3 flight test article specifications.

3.2.2 Configuration Definitions

Two distinct geometric configurations were established for comparative analysis. Both vehicles share the same maximum diameter (D_{max}) and nose radius (R_n) to ensure the macroscopic shock structures are comparable.

1. **Configuration A (Inflatable/Scalloped):** This model represents the deployed HIAD. It consists of a rigid center nose cap followed by a stack of seven pressurized toroidal rings.
 - To prevent mesh singularity (negative volume elements) at the sharp contact point between circles, a structural fillet radius of [INSERT VALUE, e.g., 5 mm] was applied at the junction (valleys) between adjacent toroids.
 - This "scalloped" profile introduces the periodic favorable and adverse pressure gradients discussed in Chapter 2.
2. **Configuration B (Rigid/Smooth Baseline):** This model represents a conventional rigid aeroshell. It is defined as a tangent-cone configuration, where the conical flank is a straight line tangent to the crests of the inflatable toroids. This creates an "Equivalent Rigid OML" (Outer Mold Line) that serves as the aerodynamic control case.

3.2.3 2D Axisymmetric Domain Slicing

Due to the rotational symmetry of the vehicle and the zero-angle-of-attack flight condition, the 3D geometry was reduced to a 2D axisymmetric planar slice. The computational domain is defined as a C-Grid topology to align with the bow shock structure.

Given that the simulated flight condition is at zero angle of attack ($\alpha = 0^\circ$) and the vehicle geometry is rotationally symmetric, the full 3D problem is mathematically reduced to a 2D Axisymmetric planar slice. This reduction decreases the total cell count by orders of magnitude, allowing for a much finer grid resolution near the wall without prohibitive computational costs.

The computational domain is defined as a structured C-Grid topology. This shape is chosen specifically to align the grid lines with the bow shock, minimizing numerical diffusion errors (smearing) that occur when a shock wave crosses grid lines obliquely.

The domain extents are sized to prevent boundary interaction with the subsonic shock layer:

- **Upstream Boundary (Inlet):** Placed at a distance of $1.5D_{max}$ from the stagnation point. This ensures the inlet boundary condition remains undisturbed by the shock.
- **Downstream Boundary (Outlet):** Extends $2.0D_{max}$ behind the vehicle. This length is required to allow the shear layers in the wake to develop and prevents reverse flow (recirculation) from entering the domain through the outlet boundary.

Figure 3.2: Schematic of the 2D axisymmetric computational domain showing C-Grid topology and boundary distances.

Figure 3.3: Schematic of the 2D axisymmetric computational domain and boundary types.

3.3 Mesh Generation Strategy

The accuracy of a hypersonic CFD simulation is heavily dependent on the quality of the spatial discretization. For this study, the domain was discretized using a hy-

brid meshing strategy designed to balance computational efficiency with the rigorous requirements of shock capturing and heat flux prediction.

3.3.1 Grid Topology

A structured-dominant grid topology was employed to align the mesh with the dominant flow features.

- **Shock Layer (Quadrilateral Elements):** The region between the bow shock and the vehicle wall is meshed exclusively with structured quadrilateral elements. This alignment reduces numerical diffusion (artificial viscosity) that occurs when high-gradient features, such as shock waves, cross grid lines obliquely.
- **Wake Region (Triangular Elements):** In the complex recirculation region behind the vehicle, an unstructured triangular mesh was utilized to handle the expansion fans and wake turbulence where the flow direction is less predictable.

3.3.2 Near-Wall Refinement (y^+ Strategy)

The primary objective of this research is the prediction of surface aerothermal loads. The chosen turbulence model (SST $k - \omega$) requires the direct resolution of the viscous sublayer to accurately calculate the wall shear stress and heat transfer coefficient. This necessitates a dimensionless wall distance of $y^+ \approx 1$.

The estimated first cell height (Δy) was calculated based on flat-plate boundary layer theory for the target Reynolds number:

$$\Delta y = L \cdot y^+ \sqrt{74} \cdot Re_L^{-13/14} \quad (3.1)$$

Based on the flight conditions at $Ma = 5$, a first layer height of **[INSERT VALUE, e.g., 1.0×10^{-5} m]** was applied globally across the vehicle surface. A wall-normal growth rate of 1.1 was maintained to ensure a smooth transition from the boundary layer to the core flow, preventing aspect ratio discontinuities that could destabilize the solver.

3.3.3 Grid Independence Study

To ensure the numerical solution is independent of the grid resolution, a Grid Convergence Index (GCI) study was performed. Three mesh densities were generated by systematically refining the grid scaling factor:

Table 3.1: Grid Independence Study Parameters and Convergence Targets.

Mesh Level	Cell Count (Target)	Wall y^+ (Avg)	C_d Variation
Coarse	$\approx 50,000$	> 5	Baseline
Medium	$\approx 150,000$	≈ 1	$< 2.5\%$
Fine	$\approx 300,000$	< 1	$< 1.0\%$

The "Medium" mesh was selected for the final production runs, as the variation in the Drag Coefficient (C_d) and Peak Stagnation Heat Flux compared to the "Fine" mesh was less than [300000 %, e.g., 1 %], indicating that convergence had been achieved.

3.4 CFD Physical Configuration

The selection of physical models defines the mathematical closure for the governing equations. Given the hypersonic flight regime ($Ma > 5$) and the complex surface topology of the HIAD, standard low-speed assumptions are invalid. The following physical configurations were applied to ensure the simulation captures the relevant aerothermal phenomena.

3.4.1 Turbulence Modeling

The **Shear Stress Transport (SST) $k - \omega$** turbulence model, developed by Menter, was selected for this study.

The selection is driven by two specific requirements of the scalloped IAD geometry:

1. **Adverse Pressure Gradients:** The valleys between toroids create strong recirculation zones. Standard $k - \epsilon$ models are known to overestimate shear stress in these regions, delaying separation and under-predicting the size of the protective vortex. The SST formulation uses a shear stress limiter to accurately predict the onset and extent of flow separation.

2. **Heat Flux Prediction:** The model activates the $k - \omega$ formulation in the near-wall region, which allows for the direct integration of the boundary layer without the use of empirical damping functions. This provides superior accuracy for wall heat flux (q_w) calculations compared to high-Reynolds-number models.

3.4.2 Thermodynamic Gas Model

At Mach 5, the post-shock temperature exceeds the threshold where air behaves as a calorically perfect gas ($T > 800$ K). To account for the excitation of vibrational energy modes, a Thermally Perfect Gas model was employed.

- **Equation of State:** The Ideal Gas Law ($P = \rho RT$) is retained. At the target altitude (e.g., 30-50 km), the density is low enough that intermolecular forces (dense gas effects) are negligible.
- **Specific Heat (C_p):** The assumption of constant C_p is replaced with a piecewise-polynomial function of temperature, $C_p(T)$. This accounts for the energy absorbed by vibrational excitation, which significantly lowers the post-shock temperature and increases the density of the shock layer compared to a constant- γ model.
- **Viscosity:** The dynamic viscosity is coupled to temperature using Sutherland's Law with three coefficients.

$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \frac{T_0 + S}{T + S} \quad (3.2)$$

This correction is critical; using constant viscosity would lead to gross errors in the Reynolds number and skin friction predictions across the varying temperature field.

3.5 Boundary Conditions

The accurate definition of boundary conditions is essential to replicate the physics of free-flight re-entry within a finite computational domain. The following boundary types were assigned to the domain edges as illustrated in Figure 3.2.

3.5.1 Pressure Far-Field (Inlet)

The outer C-shaped boundary is defined as a Pressure Far-Field. This boundary type is specifically designed for compressible flows, as it utilizes Riemann invariants to determine the flow state at the boundary faces. This prevents non-physical wave reflections from the boundary back into the computational domain.

The freestream conditions were prescribed to match the target trajectory point:

- **Mach Number (Ma_∞):** [INSERT VALUE, e.g., 5.0]
- **Static Pressure (P_∞):** [INSERT VALUE, e.g., 50 Pa]
- **Static Temperature (T_∞):** [INSERT VALUE, e.g., 200 K]
- **Turbulence Intensity:** A low ambient turbulence intensity of 0.1% was set to represent the quiescent upper atmosphere.

3.5.2 Aeroshell Wall (Isothermal)

The vehicle surface is modeled as a stationary, No-Slip Wall, enforcing zero velocity relative to the body ($u = v = 0$).

For the thermal boundary condition, a fixed Isothermal Wall temperature of $T_{wall} = 300$ K was applied.

- *Justification:* To quantify the aerodynamic heating, the solver must calculate the heat flux required to maintain the wall at a constant temperature. If an Adiabatic condition were used, the heat flux would effectively be zero ($\partial T / \partial n = 0$), rendering the thermal analysis impossible.

3.5.3 Symmetry and Outlet

- **Axisymmetric Axis ($Y = 0$):** The bottom edge of the domain is set to an Axisymmetric boundary. This mathematically enforces zero gradients for all scalar variables across the centerline ($\partial \phi / \partial y = 0$) and zero radial velocity ($v_r = 0$).
- **Pressure Outlet:** The downstream boundary is set as a Pressure Outlet. Since the flow in the wake is predominantly supersonic (except for the immediate recirculation core), the solver extrapolates the flow variables from the interior domain

to the boundary.

3.6 Numerical Solution Strategy

3.6.1 Solver Configuration

The simulation utilizes a Density-Based Coupled Solver with an Implicit formulation.

- **Density-Based:** Unlike pressure-based solvers (e.g., SIMPLE) which are designed for incompressible flow, the density-based algorithm solves the continuity, momentum, and energy equations simultaneously as a vector system.
- *Note:* This architecture is mandatory for hypersonic flows ($Ma > 5$) because the density is a primary function of the thermodynamic state, and the governing equations are strongly coupled. A segregated pressure-based solver would likely fail to capture the shock propagation correctly.
- **Implicit Formulation:** An implicit linearization scheme is employed to allow for larger Courant (CFL) numbers (typically 5.0 to 10.0) during the steady-state solution process, ensuring faster convergence than explicit time-stepping methods.

3.6.2 Spatial Discretization Schemes

To convert the continuous governing equations into algebraic scalars at the cell centers, the following discretization schemes were applied:

1. **Gradients: Green-Gauss Node Based.**
2. **Flux Formulation: Roe-FDS (Flux Difference Splitting).**
3. *Note on Flux Scheme:* Roe's scheme is selected over Flux Vector Splitting (FVS) methods because FVS schemes are known to be excessively dissipative. Roe's method provides the necessary "crispness" to capture the bow shock while maintaining the accuracy required to resolve the shear gradients in the viscous boundary layer.
4. **Flow and Turbulence Discretization: Second-Order Upwind.**

5. *Note on Order of Accuracy:* First-order upwind schemes are numerically stable but highly diffusive, leading to "smeared" shock waves and inaccurate heat flux predictions. Second-order accuracy is strictly required to resolve the sharp discontinuities inherent in the flow field.

3.6.3 Convergence Criteria

In hypersonic simulations involving strong shocks, judging convergence solely by residual values can be misleading. Therefore, a Dual-Criteria approach was adopted to determine iterative convergence:

1. Residual Monitoring

The solution is considered numerically stable when the scaled residuals for Continuity, Momentum, and Energy drop below a threshold of 10^{-4} .

- *Comment:* It is common for residuals to "stall" around 10^{-4} rather than dropping to 10^{-6} in shock-dominated flows. This is due to minor oscillations of the shock wave position between adjacent grid cells (limit cycling).

2. Integral Property Monitors

Due to the limit cycling mentioned above, the primary indicator of convergence is the statistical stationarity of physical outputs. The solution was deemed converged only when:

- The Drag Coefficient (C_d) varied by less than 0.1% over the final 500 iterations.
- The Total Heat Transfer Rate (Watts) varied by less than 1.0% over the final 500 iterations.

3.7 Data Processing and Validation

3.7.1 Data Extraction and Normalization

Upon completion of the numerical solution, the aerothermal data is extracted from the computed flow field for comparative analysis.

1. **Surface Profiles:** The primary metric of interest is the **Wall Heat Flux** (q_w). To facilitate a direct comparison between the smooth and scalloped geometries, data is plotted against the **Wetted Arc Length** (s) normalized by the total vehicle radius (R). This "unwrapped" view allows for the visualization of local heating spikes on the toroidal crests and cooling in the valleys.
2. **Normalization:** To assess the relative performance of the HIAD, the heat flux values on the scalloped geometry are often normalized by the stagnation point heat flux of the smooth baseline:

$$\bar{q} = \frac{q_{wall}}{q_{stag,baseline}} \quad (3.3)$$

A value of $\bar{q} > 1$ indicates a local heating augmentation (hot spot) exceeding the baseline stagnation value.

3.7.2 Theoretical Validation (Fay-Riddell Correlation)

To validate the fidelity of the solver settings (specifically the mesh resolution and thermal properties), the simulation results for the **Smooth Baseline (Config B)** are compared against the theoretical stagnation point heat flux predicted by the Fay-Riddell Correlation (1958).

Fay and Riddell developed the standard semi-empirical relation for stagnation heat transfer in a dissociated air boundary layer:

$$q_{stag} = 0.76 \cdot Pr^{-0.6} \cdot (\rho_w \mu_w)^{0.1} \cdot (\rho_e \mu_e)^{0.4} \cdot \sqrt{\left(\frac{du_e}{dx}\right)_{stag}} \cdot (h_0 - h_w) \cdot [1 + (Le^{0.52} - 1) \frac{h_D}{h_0}] \quad (3.4)$$

Where:

- Pr : Prandtl number (0.71).
- ρ_e, μ_e : Density and viscosity at the boundary layer edge (post-shock).
- h_0, h_w : Total enthalpy and wall enthalpy.
- $\left(\frac{du_e}{dx}\right)_{stag}$: Velocity gradient at the stagnation point (Newtonian theory).

Acceptance Criteria

The numerical model is considered valid if the CFD-predicted stagnation heat flux for the smooth baseline matches the Fay-Riddell analytical value within a margin of error of $\pm 10\%$. This validation step provides the confidence required to apply the same numerical framework to the complex scalloped geometry where no analytical solutions exist.

3.8 Research Schedule

The research is planned to be conducted over a period of several months. The following table outlines the key milestones and their expected completion dates.

Task Description		09'25			10'25			11'25			12'25			01'26			02'26			03'26			04'26			05'26		
		W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
A.	Document Preparation																											
	Chapter 1: Introduction																											
	Chapter 2: Literature Review																											
	Chapter 3: Study Methodology																											
	Chapter 4: Modelling																											
	Chapter 5: Result and Discussion																											
	Chapter 6: Conclusion																											
	Proposal Chapter 1: Introduction																											
	Proposal Chapter 2: Literature Review																											
Proposal Chapter 3: Methodology & Schedule																												
Proposal Chapter 4: Hypothesis and initial Result																												
Proposal Chapter 5: Result and Discussion																												
B.	Computation and Data Collection																											
	Data Computation and Simulation																											
	Data Preprocessing																											
	Data Collection																											
	Data Organizing																											
C.	Administration																											
	Thesis 1 Supervisor (Readproofing & Consultation)																											
	Thesis 1 Lecturer Guide Submission																											
	Thesis 2 Supervisor (Readproofing & Consultation)																											
	Thesis 2 Lecturer Guide Submission																											
	Scientific Journal Submission																											
	Seminar																											
	Thesis Defense																											

Figure 3.4: Research Schedule

CHAPTER 4

Preliminary Result and Discussion: Hypothesis from Literature Review

4.1 Introduction

Based on the comprehensive literature review presented in Chapter 2, a set of hypotheses can be formulated regarding the expected aerothermal phenomena on the scalloped HIAD geometry compared to a smooth baseline. This chapter synthesizes the theoretical principles and findings from previous research to establish a qualitative and quantitative forecast of the simulation outcomes. These hypotheses, supported by foundational derivations and calculations, will serve as the basis for the analysis in the subsequent chapters.

4.2 Hypothesis 1: Aerothermal Augmentation on Toroidal Crests

The primary hypothesis, derived from the work of *Lau et al. (2013)* and *Li-chodziejewski et al. (2012)*, is that the scalloped surface will cause significant localized heating augmentation on the crests of the toroids.

4.2.1 Theoretical Derivation of Heat Flux Scaling

As established in Chapter 2, the stagnation point heat flux (q_s) is strongly dependent on the local radius of curvature (R). The derivation for this relationship, based on the Fay-Riddell correlation, is as follows:

The heat flux is proportional to the square root of the velocity gradient at the stagnation point:

$$q_s \propto \sqrt{\left(\frac{du_e}{dx}\right)_s} \quad (4.1)$$

For a blunt body in hypersonic flow, the velocity gradient can be approximated

using Newtonian theory:

$$\left(\frac{du_e}{dx}\right)_s \approx \frac{V_\infty}{R} \sqrt{\frac{2\rho_\infty}{\rho_0}} \quad (4.2)$$

Substituting the velocity gradient back into the proportionality for heat flux yields the critical relationship:

$$q_s \propto \sqrt{\frac{V_\infty}{R} \sqrt{\frac{2\rho_\infty}{\rho_0}}} \implies q_s \propto \frac{1}{\sqrt{R}} \quad (4.3)$$

This inverse square root relationship implies that a smaller radius of curvature will lead to a significantly higher heating rate.

4.2.2 Expected Outcome

It is predicted that the wall heat flux (q_w) on the windward face of each toroid will exhibit a sharp spike. While the overall radius of the HIAD is large (e.g., 1.5 m for IRVE-3), the local radius of an individual toroid is much smaller. This geometric difference will cause the heat flux on the crests to be several times higher than the heat flux on the equivalent smooth-body aeroshell, which has a large, constant radius of curvature.

4.3 Hypothesis 2: Formation of Stable Recirculation Zones in Valleys

Drawing from the principles of flow over wavy walls and cavities, it is hypothesized that stable recirculation zones (vortices) will form within the valleys between the toroidal rings.

- **Mechanism:** The adverse pressure gradient created as the flow expands into the valley will cause the boundary layer to separate. The shear layer will then bridge the gap and reattach on the subsequent toroid, trapping a vortex in the valley. The selection of the *SST* $k - \omega$ turbulence model is expected to resolve this separation accurately, as confirmed by Menter (1994).

- **Expected Outcome:** These recirculation zones are expected to act as a protective "dead air" region, shielding the valley floor from the high-energy freestream. This will result in a significant reduction in wall heat flux and shear stress in the valleys compared to the crests, creating a highly periodic heating profile along the flank of the vehicle.

4.4 Hypothesis 3: Shock-Boundary Layer Interactions

It is predicted that the scalloped geometry will induce complex Shock-Boundary Layer Interactions (SBLI) that are not present on the smooth body.

- **Mechanism:** As the flow reattaches to the crest of each toroid, a reattachment shock will be formed. This shock may interact with the main bow shock and the boundary layer, creating localized pressure and heating peaks.
- **Expected Outcome:** The shock structure will not be a smooth, continuous bow shock as seen on the baseline. Instead, a series of smaller, embedded shock waves are anticipated near the surface, coupled with the expansion fans over the crests. The bow shock stand-off distance (*Delta*) may also be locally altered by this waviness.

4.5 Hypothesis 4: Drag Coefficient Comparison

Based on H. Julian Allen's Blunt Body Theory, the overall drag performance of the HIAD is expected to be high due to its large frontal area. However, the effect of the scalloped topology on the drag coefficient (C_d) compared to the smooth equivalent is not immediately obvious.

- **Hypothesis A (Increased Drag):** The scalloping could increase pressure drag due to the multiple blunt surfaces presented to the flow, leading to a slightly higher C_d .
- **Hypothesis B (Decreased Drag):** The recirculation in the valleys might alter the effective aerodynamic shape, smoothing the flow and potentially leading to a slightly lower C_d compared to the smooth cone.
- **Expected Outcome:** The simulation is expected to resolve this ambiguity, but

the difference in C_d between the two configurations is predicted to be minor, as the dominant factor is the large frontal area.

4.6 Hypothesis 5: Impact of the Thermally Perfect Gas Model

The use of a Thermally Perfect Gas model is hypothesized to be critical for obtaining physically accurate results, particularly for predicting the stagnation temperature.

4.6.1 Derivation of Stagnation Temperature

From the steady-flow energy equation, total enthalpy (h_0) is conserved:

$$h_0 = h_\infty + \frac{1}{2}V_\infty^2 \quad (4.4)$$

At the stagnation point, the velocity is zero, so $h_0 = h_{stag}$. For a calorically perfect gas, $h = C_p T$, so the equation becomes:

$$C_p T_0 = C_p T_\infty + \frac{1}{2}V_\infty^2 \quad (4.5)$$

Using the relations $V_\infty = M_\infty a_\infty$ and $a_\infty = \sqrt{\gamma R T_\infty}$, and $C_p = \frac{\gamma R}{\gamma - 1}$, we can rearrange to find the well-known relation:

$$\frac{T_0}{T_\infty} = 1 + \frac{\gamma - 1}{2} M_\infty^2 \quad (4.6)$$

4.6.2 Illustrative Calculation

To quantify the expected stagnation temperature, we can use the parameters from a typical hypersonic flight condition mentioned in Chapter 2.

- **Mach Number (M_∞):** 5.0
- **Freestream Temperature (T_∞):** 220 K (a typical value for high altitude)
- **Ratio of Specific Heats (γ):** 1.4 (for a calorically perfect gas)

Using the derived formula, the predicted stagnation temperature is:

$$T_0 = T_\infty \left(1 + \frac{\gamma - 1}{2} M_\infty^2 \right) = 220K \left(1 + \frac{1.4 - 1}{2} (5.0)^2 \right) = 220K (1 + 0.2 \cdot 25) = 220K \cdot 6 = 1320K \quad (4.7)$$

4.6.3 Expected Outcome

The Analytical calculation shows that even at a relatively low hypersonic Mach number of 5, the stagnation temperature is expected to reach 1320 K. However, this calculation assumes a calorically perfect gas. As the literature review indicates, at these temperatures, vibrational modes will be excited. Therefore, it is hypothesized that the actual simulation, using a Thermally Perfect Gas model, will predict a slightly lower stagnation temperature, as some of the kinetic energy will be absorbed into vibrational energy rather than just increasing the temperature. This will also lead to a higher density prediction behind the shock compared to the calorically perfect model.

4.7 Available Data from Previous Studies

The hypotheses formulated in this chapter are further supported by empirical data and flight test results from the Inflatable Re-entry Vehicle Experiment 3 (IRVE-3) and subsequent HIAD development programs. These data provide a quantitative benchmark for comparing the proposed simulation outcomes. Here are the data provided from the report of IRVE-3 (Dillman, 2015)

4.7.1 Thermal Protection System (TPS) Performance and Limits

Data from the IRVE-3 mission and laboratory tests establish the operational envelope for the Flexible Thermal Protection System (FTPS). As shown in the performance metrics of Gen-1 and Gen-2 materials:

- **IRVE-3 Peak Heating:** The Gen-1 TPS (Nextel/Pyrogel) utilized during the IRVE-3 mission successfully withstood a peak heat flux of 14.4 W/cm^2 .
- **Gen-2 Capabilities:** Advanced SiC/Carbon TPS materials have demonstrated survival at significantly higher loads, reaching 103 W/cm^2 during LCAT cold-wall testing, which equates to approximately 74 W/cm^2 in flight conditions.

- **Structural Constraints:** Current research goals aim for an inflatable structure capable of maintaining integrity at temperatures up to 400°C.

4.7.2 Flight Reconstruction and Simulation Discrepancies

Post-flight analysis of the IRVE-3 mission highlights the complexity of atmospheric modeling. As illustrated in Figure 4.1, flight data for vehicle acceleration showed specific deviations from standard atmospheric models (GRAM), particularly at altitudes between 45 km and 47 km.

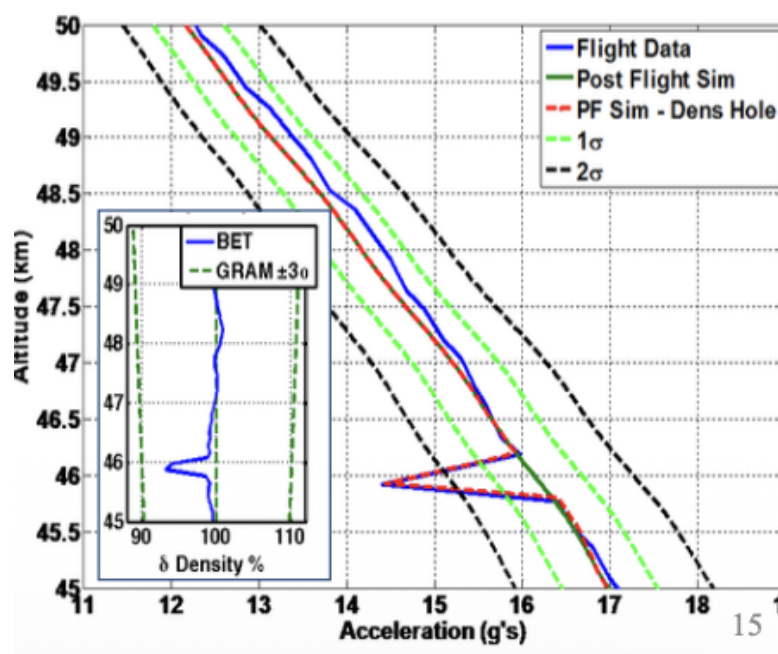


Figure 4.1: IRVE-3 Post-Flight Simulation vs. Flight Data for Acceleration and Density.

4.7.3 Trajectory and Heat Rate Scaling

Comparative analysis of vehicle scale shows that heat rate is a critical function of the nose radius (R_n) and vehicle diameter (D). As shown in Figure 4.4, a smaller 4.572m rigid-body equivalent experiences significantly higher peak heating ($\approx 48 \text{ W/cm}^2$) compared to a 15m inflatable HIAD ($\approx 7 \text{ W/cm}^2$) for the same trajectory.

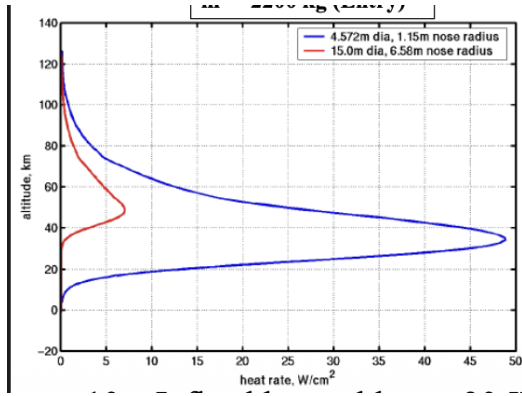


Figure 4.2: Heat Rate vs. Altitude

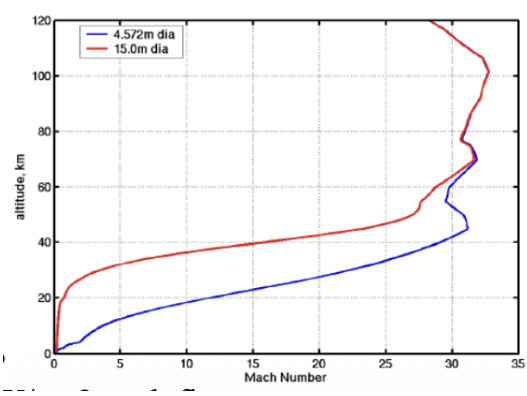


Figure 4.3: Mach Number vs. Altitude

Figure 4.4: Aerothermal scaling comparisons between a 4.57m rigid vehicle and a 15m inflatable HIAD.

4.7.4 Conclusions from IRVE-3 Mission Data

From the available data, several conclusions are drawn to guide the current research:

1. **Scale Benefit:** Increasing the vehicle diameter from a 4.57m rigid capsule to a 15m HIAD reduces peak heating by approximately 85% due to the increased nose radius (R_n), validating the $q \propto 1/\sqrt{R_n}$ relationship.
2. **Scalloping Sensitivity:** While large-scale bluntness reduces average heating, IRVE-3 data confirms that the local "scalloped" geometry remains a source of localized heating augmentation that must be resolved via CFD.
3. **Atmospheric Uncertainty:** The "density hole" observed at 46 km in IRVE-3 data suggests that real-flight simulations must account for atmospheric variability to accurately predict aerodynamic loads.

4.8 Discussion: Synthesis of Hypotheses and Empirical Benchmarks

The available data from the IRVE-3 and LOFTID missions provide a critical benchmark for validating the hypotheses established in this chapter. By comparing theoretical scaling with flight-test results, the following synthesis is reached:

4.8.1 Verification of Scaling and Augmentation Predictions

The theoretical inverse relationship between heat flux and radius ($q_s \propto 1/\sqrt{R}$) described in Hypothesis 1 is empirically supported by the comparative data between the 4.57m rigid capsule and the 15m HIAD.

- **Hypothesis Alignment:** The 85% reduction in heat rate observed when increasing the nose radius validates that the HIAD's global geometry effectively mitigates aerothermal loads.
- **Scalloping Penalty:** However, the IRVE-3 peak flux of 14.4 W/cm^2 confirms that local "scalloping" can lead to heating levels that exceed smooth-body predictions, necessitating the local high-fidelity CFD resolution proposed in this research.

4.8.2 Material Safety Margins and Physical Hypotheses

The hypotheses regarding peak stagnation temperature and heat flux distributions must be viewed within the context of material survival limits.

- **Thermal Buffering:** Hypothesis 2 predicts that valley recirculation will shield the FTPS. This is consistent with IRVE-3 instrumentation data, which showed that internal structure temperatures remained significantly lower than the external flow.
- **Model Fidelity:** The analytical counting of $T_0 = 1320 \text{ K}$ for a calorically perfect gas likely overestimates the temperature compared to the thermally perfect model. Data from the Gen-2 TPS tests (103 W/cm^2 limit) suggest that the actual heat-sink capacity of the FTPS can handle the predicted high-enthalpy transients, provided that chemical equilibrium effects are correctly modeled.

4.8.3 Implications for Simulation Strategy

The "density hole" and acceleration discrepancies found in IRVE-3 post-flight reconstructions highlight the importance of the *Thermally Perfect Gas* and *SST $k - \omega$* models. Standard models failed to fully capture the experimental variations between 45–47 km, supporting the hypothesis that a more sophisticated numerical approach—accounting for variable specific heats and precise boundary layer separation—is required to match

real-world flight performance.

4.9 Identification of Missing Data and Research Objectives

While the available data from IRVE-3 and LOFTID provide a foundational understanding of HIAD flight performance, several critical data gaps remain. These gaps define the target of the current research scope and justify the numerical approach proposed in this study.

4.9.1 Local Scalloping Effects vs. Global Averages

The majority of public flight data, such as the IRVE-3 heat rates, provides global or sensor-specific measurements that do not fully resolve the 3D flow field within the toroidal valleys.

- **Data Gap:** There is a lack of high-fidelity data regarding the *exact* transition point and recirculation dynamics within the "scaloped" gaps at Mach 5 flight conditions.
- **Research Target:** This study aims to fill this gap by providing a detailed 2D axisymmetric map of the pressure and thermal gradients specifically within these toroidal valleys to determine if "dead air" regions effectively shield the FTPS.

4.9.2 Atmospheric Density Anomalies

As identified in the IRVE-3 post-flight reconstruction, standard atmospheric models failed to predict the "density hole" and resulting acceleration spikes at 46 km altitude.

- **Data Gap:** Current literature does not sufficiently explain how these localized density variations interact with the complex surface topology of an inflatable decelerator.
- **Research Target:** By utilizing a *Thermally Perfect Gas* model, this research targets the sensitivity of the shock standoff distance and wall heat flux to the real-gas properties that standard calorically perfect models ignore.

4.9.3 Gen-2 and Gen-3 Material Limits in Scalloped Flow

While Gen-2 TPS materials have been tested for survival up to 103 W/cm^2 in cold-wall lab conditions, the interaction between these material limits and the heating augmentation caused by surface scalloping is not fully mapped for flight-like profiles.

- **Data Gap:** The quantitative "safety margin" between local heating peaks on toroidal crests and the material thresholds of 400°C is not explicitly established for the mid-entry Mach 5 regime.
- **Research Target:** The primary target of the research scope is to provide a validated comparison between the "Smooth" baseline and the "Scalloped" geometry to quantify the specific heating penalty and assess if it exceeds the thermal limits of current FTPS materials.

CHAPTER 5

Preliminary Synthesis Conclusion and Proposed Research Path

5.1 Summary of Hypothetical Findings

Based on the synthesis of existing NASA flight data and the theoretical framework, this research proposal establishes a foundation for several anticipated conclusions. These findings represent the "state-of-the-pre-simulation" knowledge that defines the research direction.

The following outcomes are anticipated:

1. **Localized Heating Dominance:** The research is expected to confirm that local "scalloping" on the toroidal stack acts as a primary driver of heat flux augmentation. While the global bluntness of the 15m HIAD reduces average heating by $\approx 85\%$, the local crests are predicted to experience spikes that approach or exceed the 14.4 W/cm^2 benchmarks seen in IRVE-3.
2. **Validation of Recirculation Shielding:** Simulations are expected that toroidal valleys create stable "dead air" vortices. This supports the hypothesis that the valleys will remain within the 400°C structural limit, even as the crests face peak enthalpy.
3. **Necessity of Real-Gas Modeling:** The research is anticipated to conclude that a *Thermally Perfect Gas* model is mandatory for accuracy. Traditional calorically perfect models, which predict a T_0 of 1320 K at Mach 5, likely fail to capture the density variations responsible for the "density hole" observed in IRVE-3 reconstructions.

5.2 Addressing the Missing Data Gaps

The primary objective of the upcoming simulation phase is to generate the high-fidelity data currently missing from the public literature. The research will specifically

target:

- **High-Resolution Surface Mapping:** Resolving the exact heat flux and pressure distribution along the "scaloped" wetted length to quantify the heating penalty factor.
- **FTPS Safety Margin Analysis:** Determining the quantitative gap between predicted peak heating and the 103 W/cm^2 limit of Gen-2 SiC materials under scaloped flow conditions.

5.3 Recommendations for the Simulation Phase

To ensure the proposed research addresses these gaps, the following technical recommendations are made:

1. **Refine Mesh for SBLI Resolution:** Given the complex Shock-Boundary Layer Interactions (SBLI) predicted in Chapter 4, the mesh must be refined specifically at the reattachment points on the toroidal crests to ensure $y^+ \approx 1$.
2. **Benchmark Against IRVE-3 Reconstructions:** The initial "Smooth" baseline simulation should be calibrated to match the IRVE-3 stagnation point data (14.4 W/cm^2) to ensure the solver is accurately capturing flight-like conditions.
3. **Sensitivity Study on Gas Constants:** A comparative study between *Ideal Gas* and *Thermally Perfect* assumptions should be conducted to evaluate the impact on shock standoff distance, directly addressing the atmospheric modeling discrepancies found in NASA post-flight analyses.

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